

code_dont_fail_me_now

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1	Templates and Scripts	
1.1	sublime	
	{	
	"shell_cmd": "g++ -Wall -std=c++20 -O2	
	-DLOCAL \$file -o \$file_base_name	
	&& bash -c 'ulimit -S -f 10000;	
	timeout 5 ./ \$file_base_name < input	
	.in > output.out 2> error.log',	
	"file_regex": "~(..[~:]*):([0-9]+)	
	:?([0-9]+)??.*\$',	
	"working_dir": "\${file_path}",	
	"selector": "source.cpp",	
	// sanitize flags: -fsanitize=address	
	-fsanitize=undefined	
	}	
1.2	vs code	
	{	
	"version": "2.0.0",	
	"tasks": [{	
	"label": "cp",	
	"type": "shell",	
	"command": "g++ -Wall -std=c++20 -O2	
	-DLOCAL \"\${file}\" -o \"\${	
	fileDirname}/\${	
	fileBasenameNoExtension}\" && \"\${	
	fileDirname}/\${	
	fileBasenameNoExtension}\" < input.	
	in > output.out 2> error.log",	
	"group": "build",	

```
"presentation": { "reveal": "silent"
},
"problemMatcher": ["$gcc"]
}]
}
```

1.3 Interactor

```
socat EXEC:"./interactor" EXEC:"./soln"
```

1.4 debug.h

```
#include <iostream>
using namespace std;
#define see(x) cerr << #x << ": " << x
<< nl
template <typename... Args>
void debug(Args... args) {
    ((cerr << " " << args), ...) << "\n";
}
```

2 Number Theory

2.1 mini mod

```
inline ll mod(ll n, ll m = MOD) { return
    (m + n % m) % m; }
ll modpow(ll n, ll k, ll m = MOD) {
    ll ret = 1;
    n = mod(n, m);
    while (k) {
        if (k & 1) ret = (ret * n) % m;
        n = (n * n) % m;
        k >>= 1;
    }
    return ret;
}
inline ll inv(ll n, ll m = MOD) { return
    modpow(n, m - 2, m); }
inline ll mul(ll x, ll y, ll m = MOD) {
    return (mod(x, m) * mod(y, m)) % m;
}
inline ll dvd(ll x, ll y, ll m = MOD) {
    return mul(x, inv(y, m), m); }
```

2.2 Binomial Coefficient

```
ll fact[MAXN];
void calc_fact(ll m = MOD) {
    fact[0] = 1;
    for (int i = 1; i < MAXN; ++i) {
        fact[i] = (fact[i - 1] * i) % m;
    }
}
ll C(int n, int k, ll m = MOD) {
    if (k > n) return 0;
    return dvd(fact[n], fact[k] * fact[n -
        k] % m, m);
}
```

2.3 Sieve, SPF, Phi

```
vector<int> primes;
bool isprime[MAXN];
int spf[MAXN], phi[MAXN];
void sieve() {
    for (int i = 0; i < MAXN; ++i) {
        isprime[i] = true; spf[i] = i; phi
        [i] = i;
    }
    isprime[0] = isprime[1] = false;
    for (ll i = 2; i < MAXN; ++i) {
        if (isprime[i]) {
            primes.push_back(i); phi[i] =
            i - 1;
            for (ll j = 2 * i; j < MAXN; j
            += i) {
                isprime[j] = false;
                if (spf[j] == j) spf[j] =
                i;
            }
        }
    }
}
```

```

        phi[j] -= phi[j] / i;
    }
}
}
}

```

2.4 Segmented Sieve

```

// Generate all primes from l to r using
// segmented sieve in
// O((r - l) log (r) + sqrt(r)) // needs
// sieve(limit)
vector<int64_t> segmented_sieve(int64_t
    l, int64_t r) {
    if (l == 1) l++;
    int limit = sqrtl(r);
    while (limit * limit <= r) limit++;
    while (limit * limit > r) limit--;
    auto primes = sieve(limit);
    vector<bool> is_prime(r - l + 1, true);
    for (int64_t p : primes) {
        int64_t start = max(p * p, (l + p -
            1) / p * p);
        for (int64_t j = start; j <= r; j +=
            p) {
            is_prime[j - l] = false;
        }
    }
    vector<int64_t> vec;
    for (int64_t i = l; i <= r; ++i) {
        if (is_prime[i - l]) vec.push_back(i);
    }
    return vec;
}

```

2.5 Safe LCM

```

__int128 safe_lcm(__int128 a, __int128 b
    , __int128 lim) {
    __int128 t = ((a) / gcd(a, b));
    if (t > (lim / b)) return lim+1;
    else return t * b;
}

```

2.6 Extended GCD

```

int extended_gcd(ll a, ll b, ll& x, ll&
    y) {
    if (!b) {
        x = 1, y = 0;
        return a;
    }
    ll g = extended_gcd(b, a % b, x, y);
    swap(x, y);
    y -= x * (a / b);
    return g;
}

```

2.7 CRT

```

ll modinv(ll a, ll m) {
    ll x, y;
    extended_gcd(a, m, x, y);
    x %= m;
    return x < 0 ? x + m : x;
}
ll crt(vector<pair<ll, ll>>& mods) {
    ll mod = 1;
    for (auto& p : mods) mod *= p.ss;
    ll ans = 0;
    for (auto [x, p] : mods) {
        ll m = mod / p;
        m = (m * modinv(m, p)) % mod;
        m = (m * x) % mod;
        ans = (ans + m) % mod;
    }
    return ans;
}

```

2.8 Diophantine Equation

```

// Solves a*x + b*y = c
bool find_any_solution(ll a, ll b, ll c, ll
    &x0, ll &y0, ll &g){

```

```

    if(a==0&&b==0){
        if(c) return false;
        x0=y0=g=0; return true;
    }
    g=extended_gcd(abs(a),abs(b),x0,y0);
    if(c%g!=0) return false;
    x0*=c/g; y0*=c/g;
    if(a<0) x0*=-1;
    if(b<0) y0*=-1;
    return true;
}
// Shift solution by cnt
void shift_solution(ll &x, ll &y, ll a, ll
    b, ll cnt){
    x+=cnt*b; y-=cnt*a;
}
pair<ll, vector<pair<ll, ll>>>
    find_all_solutions(ll a, ll b, ll c,
        ll minx, ll maxx, ll
        miny, ll maxy, bool solves){
    ll x, y, g;
    if(!find_any_solution(a,b,c,x,y,g))
        return {0, {}};
    if(a==0&&b==0){
        assert(c==0);
        ll cnt = 1LL*(maxx-minx+1)*(maxy
            -miny+1);
        vector<pair<ll, ll>> res;
        if(solves){
            for(ll cur_x=minx; cur_x<=
                maxx; ++cur_x){
                for(ll cur_y=miny; cur_y
                    <=maxy; ++cur_y){
                    res.push_back({cur_x
                        , cur_y});
                }
            }
            return {cnt, res};
        }
        if(a==0){
            ll cnt = (maxx-minx+1)*(miny<=c/
                b&&c/b<=maxy);
            vector<pair<ll, ll>> res;
            if(solves && cnt > 0){
                ll cur_y = c/b;
                for(ll cur_x=minx; cur_x<=
                    maxx; ++cur_x){
                    res.push_back({cur_x,
                        cur_y});
                }
            }
            return {cnt, res};
        }
        if(b==0){
            ll cnt = (maxy-miny+1)*(minx<=c/
                a&&c/a<=maxx);
            vector<pair<ll, ll>> res;
            if(solves && cnt > 0){
                ll cur_x = c/a;
                for(ll cur_y=miny; cur_y<=
                    maxy; ++cur_y){
                    res.push_back({cur_x,
                        cur_y});
                }
            }
            return {cnt, res};
        }
        ll orig_a = a, orig_b = b;
        a/=g; b/=g;
        ll sign_a=a>0?1:-1, sign_b=b>0?1:-1;
        shift_solution(x,y,a,b,(minx-x)/b);
        if(x<minx) shift_solution(x,y,a,b,
            sign_b);
        if(x>maxx) return {0, {}};
        ll lx1=x;
        shift_solution(x,y,a,b,(maxx-x)/b);
        if(x>maxx) shift_solution(x,y,a,b,-
            sign_b);
        ll rx1=x;
        shift_solution(x,y,a,b,-(miny-y)/a);
        if(y<miny) shift_solution(x,y,a,b,-
            sign_a);
        if(y>maxy) return {0, {}};
        ll lx2=x;
        shift_solution(x,y,a,b,-(maxy-y)/a);
        if(y>maxy) shift_solution(x,y,a,b,
            sign_a);

```

```

        ll rx2=x;
        if(lx2>rx2) swap(lx2,rx2);
        ll lx=max(lx1,lx2), rx=min(rx1,rx2);
        if(lx>rx) return {0, {}};
        ll cnt = (rx-lx)/abs(b)+1;
        vector<pair<ll, ll>> res;
        if(solves){
            for(ll cur_x=lx; cur_x<=rx;
                cur_x+=abs(b)){
                res.push_back({cur_x, (c-
                    orig_a*cur_x)/orig_b});
            }
        }
        return {cnt, res};
}

```

2.9 Pollar Rho Factorization

```

namespace PollardRho {
    mt19937 rnd(chrono::steady_clock::
        now().time_since_epoch().count());
    const int P = 1e6 + 9;
    ll seq[P];
    int primes[P], spf[P];
    inline ll add_mod(ll x, ll y, ll m)
    {
        return (x += y) < m ? x : x - m;
    }
    inline ll mul_mod(ll x, ll y, ll m)
    {
        ll res = (ll)((__int128)x * y %
            m);
        return res;
    }
    inline ll pow_mod(ll x, ll n, ll m)
    {
        ll res = 1 % m;
        for (; n >= 1) {
            if (n & 1) res = mul_mod(res
                , x, m);
            x = mul_mod(x, x, m);
        }
        return res;
    }
    // O(it * (logn)^3), guaranteed
    // accurate for all 64-bit integers
    inline bool miller_rabin(ll n) {
        if (n <= 2 || (n & 1 ^ 1))
            return (n == 2);
        if (n < P) return spf[n] == n;
        ll c, d, s = 0, r = n - 1;
        for (; !(r & 1); r >>= 1, s++)
            ;
        // Bound shifted to < 40 to
        // include prime base 37
        for (int i = 0; primes[i] < n &&
            primes[i] < 40; i++) {
            c = pow_mod(primes[i], r, n)
                ;
            for (int j = 0; j < s; j++)
                {
                    d = mul_mod(c, c, n);
                    if (d == 1 && c != 1 &&
                        c != (n - 1)) return false;
                    c = d;
                }
            if (c != 1) return false;
        }
        return true;
    }
    void init() {
        int cnt = 0;
        for (int i = 2; i < P; i++) {
            if (!spf[i]) primes[cnt++] =
                spf[i] = i;
            for (int j = 0, k; (k = i *
                primes[j]) < P; j++) {
                spf[k] = primes[j];
                if (spf[i] == spf[k])
                    break;
            }
        }
    }
    // O(n^(1/4))
    ll pollard_rho(ll n) {
        while (1) {
            ll x = rnd() % n, y = x, c =

```

```

    rnd() % n, u = 1, v, t = 0;
    ll *px = seq, *py = seq;
    while (1) {
        *py++ = y = add_mod(
mul_mod(y, y, n), c, n);
        *py++ = y = add_mod(
mul_mod(y, y, n), c, n);
        if ((x = *px++) == y)
break;
        v = u;
        // Direct ternary avoids
        potential 32-bit truncation from
        global abs()
        u = mul_mod(u, (y > x ?
y - x : x - y), n);
        if (!u) return __gcd(v,
n);
        if (++t == 32) {
            t = 0;
            if ((u = __gcd(u, n)
) > 1 && u < n) return u;
        }
        if (t && (u = __gcd(u, n)) >
1 && u < n) return u;
    }
}
vector<ll> factorize(ll n) {
    if (n == 1) return vector<ll>();
    if (miller_rabin(n)) return
vector<ll>{n};
    vector<ll> v, w;
    while (n > 1 && n < P) {
        v.push_back(spf[n]);
        n /= spf[n];
    }
    if (n >= P) {
        ll x = pollard_rho(n);
        v = factorize(x);
        w = factorize(n / x);
        v.insert(v.end(), w.begin(),
w.end());
    }
    return v;
}
}

```

3 Dynamic Programming

3.1 SOS DP

```

const int B = 20;
int a[1 << B], f[1 << B], g[1 << B];
for (int i = 1; i <= n; i++) {
    cin >> a[i];
    f[a[i]]++;
    g[a[i]]++;
}
// sum over subsets
for (int i = 0; i < B; i++) {
    for (int mask = 0; mask < (1 << B);
mask++) {
        if ((mask & (1 << i)) != 0) {
            f[mask] += f[mask ^ (1 << i)];
        }
    }
}
// sum over supersets
for (int i = 0; i < B; i++) {
    for (int mask = (1 << B) - 1; mask >=
0; mask--) {
        if ((mask & (1 << i)) == 0) g[mask]
+= g[mask ^ (1 << i)];
    }
}
}

```

3.2 CHT

```

struct CHT {
    vector<ll> m, b;
    int ptr = 0;
    bool bad(int l1, int l2, int l3) {
        return 1.0 * (b[l3] - b[l1]) * (m[l1]
- m[l2]) <=
1.0 * (b[l2] - b[l1]) *

```

```

(m[l1] - m[l3]); // (
slope dec+query min), (slope inc+
query max)
return 1.0 * (b[l3] - b[l1]) * (m[l1]
- m[l2]) >
1.0 * (b[l2] - b[l1]) *
(m[l1] - m[l3]); // (
slope dec+query max), (slope inc+
query min)}
void add(ll _m, ll _b) {
    m.push_back(_m);
    b.push_back(_b);
    int s = m.size();
    while (s >= 3 && bad(s - 3, s - 2,
s - 1)) {
        s--;
        m.erase(m.end() - 2);
        b.erase(b.end() - 2);
    }
}
ll f(int i, ll x) { return m[i] * x
+ b[i]; }
// (slope dec+query min), (slope inc+
query max) -> x increasing
// (slope dec+query max), (slope inc+
query min) -> x decreasing
ll query(ll x) {
    if (ptr >= m.size()) ptr = m.size
() - 1;
    while (ptr < m.size() - 1 && f(ptr
+ 1, x) < f(ptr, x)) ptr++;
    return f(ptr, x);
}
ll bs(int l, int r, ll x) {
    int mid = (l + r) / 2;
    if (mid + 1 < m.size() && f(mid +
1, x) < f(mid, x))
        return bs(mid + 1, r, x); // >
for max
    if (mid - 1 >= 0 && f(mid - 1, x)
< f(mid, x))
        return bs(l, mid - 1, x); // >
for max
    return f(mid, x);
}
}

```

3.3 CHT Dynamic

```

struct Line {
    mutable int64_t m, b, p;
    bool operator<(const Line& o) const {
        return m < o.m; }
    bool operator<(int64_t x) const {
        return p < x; }
};
struct CHT : multiset<Line, less<>> {
    bool isMax;
    CHT(bool isMax = true) { isMax =
isMax; }
    static int64_t div(int64_t a, int64_t
b) {
        return a / b - ((a ^ b) < 0 && a % b
);
    }
    bool isect(iterator x, iterator y) {
        if (y == end()) {
            x->p = LLONG_MAX;
            return false;
        }
        if (x->m == y->m) {
            x->p = (x->b > y->b ? LLONG_MAX :
LLONG_MIN);
        } else {
            x->p = div(y->b - x->b, x->m - y->
m);
        }
        return x->p >= y->p;
    }
}
pair<int64_t, int64_t> convert(int64_t
m, int64_t b) {
    if (isMax) return {m, b};
    return {-m, -b};
}
void add(int64_t m, int64_t b) {
    tie(m, b) = convert(m, b);
    auto z = insert({m, b, 0});

```

```

    auto y = z++;
    auto x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) {
        isect(x, y = erase(y));
    }
    while ((y = x) != begin() && (--x)->
p >= y->p) {
        isect(x, erase(y));
    }
}
static int64_t eval(const Line& l,
int64_t x) {
    return (int64_t)((__int128)l.m * x +
l.b);
}
int64_t query(int64_t x) {
    assert(!empty());
    auto l = *lower_bound(x);
    int64_t ans = eval(l, x);
    if (!isMax) ans = -ans;
    return ans;
}
};

```

4 Data Structure

4.1 Fenwick Tree

```

struct fenwick {
    int n;
    vector<ll> tree;
    fenwick(int s) {
        n = s;
        tree.assign(n + 1, 0);
    }
    ll get(int i) {
        ll res = 0;
        for (; i > 0; i -= i & -i) res +=
tree[i];
        return res;
    }
    void add(int i, ll x) {
        for (; i <= n; i += i & -i) tree[i]
+= x;
    }
    void add(int l, int r, ll x) {
        add(l, x);
        if (r < n) add(r + 1, -x);
    }
    ll get(int l, int r) { return get(r) -
get(l - 1); }
};

```

4.2 Fenwick 2D (Compressed)

```

struct FT2 {
    vector<vector<ll>> ys;
    vector<FT> ft;
    FT2(int limx) : ys(limx + 1) {}
    void fakeUpdate(int x, int y) {
        for (; x < ys.size(); x += x & -x) {
            ys[x].push_back(y);
        }
    }
    void init() {
        ft.clear();
        ft.resize(ys.size(), FT(0));
        for (int i = 0; i < ys.size(); ++i)
            sort(ys[i].begin(), ys[i].end());
            ys[i].erase(unique(ys[i].begin(),
ys[i].end()), ys[i].end());
            ft[i] = FT(ys[i].size());
        }
    }
    int update(int x, int y) {
        return lower_bound(ys[x].begin(), ys
[x].end(), y - ys[x].begin()) + 1;
    }
    int query(int x, int y) {
        return upper_bound(ys[x].begin(), ys
[x].end(), y) - ys[x].begin();
    }
    void update(int x, int y, ll dif) {
        for (; x < ys.size(); x += x & -x)

```

```

    ft[x].update(update(x, y), dif);
}
ll query(int x, int y) {
    ll sum = 0;
    for (; x > 0; x -= x & -x) {
        sum += ft[x].query(query(x, y));
    }
    return sum;
}
ll query(ll x1, ll y1, ll x2, ll y2) {
    return (query(x2, y2) - query(x1-1,
        y2) - query(x2, y1-1) + query(x1
        -1, y1-1)) % MOD;
}
};

```

4.3 BIT 2D

```

template<typename T>
struct PrefixSum2D {
    vector<vector<T>>> p;
    int rows, cols;
    T combine(T x, T y) { return x + y; }
    void build(vector<vector<T>>& a) {
        rows = a.size(); cols = a[0].size();
        p.assign(rows + 1, vector<T>(cols +
            1, 0));
        for(int i = 1; i <= rows; ++i)
            for(int j = 1; j <= cols; ++j)
                p[i][j] = a[i-1][j-1] + p[i-1][j]
                    + p[i][j-1] - p[i-1][j-1];
    }
    // query sum over [r1,c1]..[r2,c2] (1-
    // indexed)
    T query(int r1, int c1, int r2, int c2) {
        return p[r2][c2] - p[r1-1][c2] - p[
            r2][c1-1] + p[r1-1][c1-1];
    }
};

```

4.4 Segment Tree

```

template <typename T> // 1 based
struct ST {
#define lc (nd << 1)
#define rc ((nd << 1) | 1)
    vector<T> t;
    T identity = 0; // 0 for sum, 1e18
    for min etc;
    ST(int n) { t.resize(4 * n, identity); }
    inline T combine(T a, T b) { return a
        + b; }
    inline void pull(int nd) { t[nd] =
        combine(t[lc], t[rc]); }
    void build(int nd, int b, int e,
        vector<T>& a) {
        if (b == e) {
            t[nd] = a[b];
            return;
        }
        int mid = (b + e) >> 1;
        build(lc, b, mid, a);
        build(rc, mid + 1, e, a);
        pull(nd);
    }
    void upd(int nd, int b, int e, int i,
        T v) {
        if (b == e) {
            t[nd] = v; // For "add v", use t[
            nd] += v;
            return;
        }
        int mid = (b + e) >> 1;
        if (i <= mid) upd(lc, b, mid, i, v);
        else upd(rc, mid + 1, e, i, v);
        pull(nd);
    }
    T query(int nd, int b, int e, int i,
        int j) {
        if (i > e || b > j) return identity;
        if (i <= b && e <= j) return t[nd];
        int mid = (b + e) >> 1;

```

```

        return combine(query(lc, b, mid, i,
            j), query(rc, mid + 1, e, i, j));
    }
};

```

4.5 Lazy Propagation

```

struct ST {
#define lc (n << 1)
#define rc ((n << 1) | 1)
    // long long t[4 * N], lazy[4 * N];
    vector<ll> a, t, lazy;
    ST(vll& a) : a(a) {
        lazy.resize(4*a.size()+1), t.resize
            (4*a.size()+1);
    }
    ST(int n) {
        a.resize(n), lazy.resize(4*n), t.
            resize(4*n);
    }
    inline void push(int n, int b, int e)
    {
        if (lazy[n] == 0) return;
        t[n] = (ll) t[n] + (ll) lazy[n] * (
            ll) (e - b + 1);
        t[n] %= mod;
        if (b != e) {
            lazy[lc] = (ll) lazy[lc] + lazy[n]
                ;
            lazy[rc] = (ll) lazy[rc] + lazy[n]
                ;
            lazy[rc] %= mod, lazy[lc] %= mod;
        }
        lazy[n] = 0;
    }
    inline long long combine(long long a,
        long long b) {
        return (a + b) % mod;
    }
    inline void pull(int n) {
        t[n] = (t[lc] + t[rc]) % mod;
    }
    void build(int n, int b, int e) {
        lazy[n] = 0;
        if (b == e) {
            t[n] = a[b];
            return;
        }
        int mid = (b + e) >> 1;
        build(lc, b, mid);
        build(rc, mid + 1, e);
        pull(n);
    }
    void upd(int n, int b, int e, int i,
        int j, long long v) {
        push(n, b, e);
        if (j < b || e < i) return;
        if (i <= b && e <= j) {
            lazy[n] = v; //set lazy
            push(n, b, e);
            return;
        }
        int mid = (b + e) >> 1;
        upd(lc, b, mid, i, j, v);
        upd(rc, mid + 1, e, i, j, v);
        pull(n);
    }
    long long query(int n, int b, int e,
        int i, int j) {
        push(n, b, e);
        if (i > e || b > j) return 0; //
        return null
        if (i <= b && e <= j) return t[n];
        int mid = (b + e) >> 1;
        return (combine(query(lc, b, mid, i,
            j), query(rc, mid + 1, e, i, j)))
            % mod;
    }
};

```

4.6 Persistent Segment Tree

```

template <typename T>
struct PST {
    struct node {
        T val;
        int l, r;

```

```

    };
    int n;
    T identity = 0; // 0 for sum, INF for
    min etc
    vector<node> nodes;
    vector<int> version;
    T merge(T x, T y) { return x + y; }
    PST() {}
    PST(int sz) {
        n = sz;
        build(vector<T>(n, identity));
    }
    PST(vector<T>& a) {
        n = a.size();
        build(a);
    }
    int build(const vector<T>& a) {
        nodes.reserve(20 * n);
        int root = _build(0, n - 1, a);
        version.push_back(root);
        return version.size() - 1;
    }
    int _build(int l, int r, const vector<
        T>& a) {
        int root = nodes.size();
        nodes.push_back({});
        if (l == r) {
            nodes[root].val = a[l];
            return root;
        }
        int mid = (l + r) / 2;
        int ch = _build(l, mid, a);
        nodes[root].l = ch;
        ch = _build(mid + 1, r, a);
        nodes[root].r = ch;
        nodes[root].val = merge(nodes[nodes[
            root].l].val, nodes[nodes[root].r].
            val);
        return root;
    }
    int update(int i, T x, int v = -1) {
        if (v == -1) v = version.size() - 1;
        int root = _update(version[v], 0, n
            - 1, i, x);
        version.push_back(root);
        return version.size() - 1;
    }
    int _update(int u, int l, int r, int i
        , T x) {
        int root = nodes.size();
        node old = nodes[u];
        nodes.push_back(old);
        if (l == r) {
            nodes[root].val = x; // change =
            to += for add
            return root;
        }
        int mid = (l + r) / 2;
        if (i <= mid) {
            int ch = _update(nodes[u].l, l,
                mid, i, x);
            nodes[root].l = ch;
        }
        else {
            int ch = _update(nodes[u].r, mid +
                1, r, i, x);
            nodes[root].r = ch;
        }
        nodes[root].val = merge(nodes[nodes[
            root].l].val, nodes[nodes[root].r].
            val);
        return root;
    }
    int update_batch(const vector<pair<int
        , T>>& up, int v = -1) {
        if (v == -1) v = version.size() - 1;
        if (up.empty()) { version.push_back(
            version[v]); return version.size()
            - 1; }
        int root = _update_batch(version[v],
            0, n - 1, up, 0, (int)up.size() -
            1);
        version.push_back(root);
        return version.size() - 1;
    }
    // Updates values up[x].first with
    // delta up[x].second. Assumes that up
    // is sorted.
    int _update_batch(int u, int l, int r,

```

```

    const vector<pair<int, T>>& up,
    int ql, int qr) {
if (ql > qr) return u;
int root = nodes.size();
nodes.push_back(nodes[u]);
if (l == r) {
    for (int i = ql; i <= qr; ++i)
        nodes[root].val += up[i].second;
    return root;
}
int mid = (l + r) / 2;
int sp = upper_bound(up.begin() + ql,
    up.begin() + qr + 1, mid,
    [](int m, const pair<int, T>& p) {
        return m < p.first; }) - up.begin()
    );
nodes[root].l = _update_batch(nodes[
    u].l, l, mid, up, ql, sp - 1);
nodes[root].r = _update_batch(nodes[
    u].r, mid + 1, r, up, sp, qr);
nodes[root].val = merge(nodes[nodes[
    root].l].val, nodes[nodes[root].r].
    val);
return root;
}
T get(int l, int r, int v) { return
    _get(version[v], 0, n - 1, l, r); }
T _get(int u, int tl, int tr, int l,
    int r) {
    if (l > tr || r < tl) return
        identity;
    if (l <= tl && r >= tr) return nodes
        [u].val;
    int mid = (tl + tr) / 2;
    return merge(_get(nodes[u].l, tl,
        mid, l, r),
        _get(nodes[u].r, mid + 1, tr, l, r
        ));
}
// version is 1 indexed (0 is the
// initial before updates)
T get_kth(int ver_l, int ver_r, int k)
{
    return _get_kth(version[ver_l - 1],
        version[ver_r], 0, n - 1, k);
}
int _get_kth(int u, int v, int l, int
    r, int k) {
    if (l == r) return l;
    int mid = (l + r) / 2;
    T kl = nodes[nodes[v].l].val - nodes
        [nodes[u].l].val;
    if (kl >= k)
        return _get_kth(nodes[u].l, nodes[
            v].l, l, mid, k);
    else
        return _get_kth(nodes[u].r, nodes[
            v].r, mid + 1, r, k - kl);
}
};

```

4.7 Merge Sort Tree

```

// 0-based index
template <typename T>
struct MST {
    int n;
    vector<vector<T>> tree;
    MST(vector<T>& a) : n(a.size()) {
        tree.resize(4 * n);
        build(0, 0, n - 1, a);
    }
    void build(int u, int l, int r, vector
        <T>& a) {
        if (l == r) {
            tree[u].pb(a[l]);
            return;
        }
        int mid = (l + r) / 2;
        build(2 * u + 1, l, mid, a);
        build(2 * u + 2, mid + 1, r, a);
        tree[u].resize(r - l + 1);
        merge(all(tree[2 * u + 1]), all(tree
            [2 * u + 2]), tree[u].begin());
    }
    // Returns number of elements smaller
    // than x in range l, r

```

```

int query(int l, int r, T x) { return
    get(0, 0, n - 1, l, r, x); }
int get(int u, int tl, int tr, int l,
    int r, T x) {
    if (tl >= l && tr <= r)
        return lower_bound(all(tree[u]), x
        ) - tree[u].begin();
    if (tl > r || tr < l) return 0;
    int mid = (tl + tr) / 2;
    return get(2 * u + 1, tl, mid, l, r,
        x) +
        get(2 * u + 2, mid + 1, tr, l
        , r, x);
}
};

```

4.8 Merse Sort Tree with Update

```

// 0 based indexing
template <typename T>
struct MST {
    int n;
    vector<ordered_set<pair<T, int>>> tree
        ;
    vector<T> a;
    MST(vector<T>& v) : a(v) {
        n = v.size();
        tree.resize(4 * n);
        build(0, 0, n - 1, a);
    }
    void merge(int u, int l, int r) {
        tree[u] = tree[l];
        for (auto x : tree[r]) tree[u].
            insert(x);
    }
    void build(int u, int l, int r, vector
        <T>& a) {
        if (l == r) {
            tree[u].insert({a[l], 1});
            return;
        }
        int mid = (l + r) / 2;
        build(2 * u + 1, l, mid, a);
        build(2 * u + 2, mid + 1, r, a);
        merge(u, 2 * u + 1, 2 * u + 2);
    }
    // Returns number of elements smaller
    // than x in range l, r
    int query(int l, int r, T x) { return
        get(0, 0, n - 1, l, r, x); }
    void update(int i, T x) {
        _update(0, 0, n - 1, i, x);
        a[i] = x;
    }
    int get(int u, int tl, int tr, int l,
        int r, T x) {
        if (tl >= l && tr <= r) return tree[
            u].order_of_key({x, 0});
        if (tl > r || tr < l) return 0;
        int mid = (tl + tr) / 2;
        return get(2 * u + 1, tl, mid, l, r,
            x) +
            get(2 * u + 2, mid + 1, tr, l
            , r, x);
    }
    void _update(int u, int l, int r, int
        i, T x) {
        tree[u].erase({a[i], i});
        tree[u].insert({x, i});
        if (l == r) return;
        int mid = (l + r) / 2;
        if (i <= mid)
            _update(2 * u + 1, l, mid, i, x);
        else
            _update(2 * u + 2, mid + 1, r, i,
                x);
    }
};

```

4.9 Treap

```

struct ImplicitTreap {
    struct Node { int l, r, sz, sum, val,
        priority, lazy; };
    vector<Node> tree;
    mt19937 rnd;
    ImplicitTreap(int max_nodes = 0) {

```

```

        rnd.seed(chrono::steady_clock::now()
            .time_since_epoch().count());
        if (max_nodes > 0) tree.reserve(
            max_nodes);
    }
    int getSize(int t) { return t == -1 ?
        0 : tree[t].sz; }
    int getSum(int t) { return t == -1 ? 0
        : tree[t].sum; }
    void applyLazy(int t, int add) {
        if (t == -1) return;
        tree[t].val += add; tree[t].sum +=
            tree[t].sz * add; tree[t].lazy +=
            add;
    }
    void pull(int t) {
        if (t == -1) return;
        tree[t].sz = 1 + getSize(tree[t].l)
            + getSize(tree[t].r);
        tree[t].sum = tree[t].val + getSum(
            tree[t].l) + getSum(tree[t].r);
    }
    void push(int t) {
        if (t == -1 || tree[t].lazy == 0)
            return;
        applyLazy(tree[t].l, tree[t].lazy);
        applyLazy(tree[t].r, tree[t].lazy);
        tree[t].lazy = 0;
    }
    int newNode(int val) {
        tree.push_back({-1, -1, 1, val, val,
            (int)rnd(), 0});
        return tree.size() - 1;
    }
    pair<int, int> split(int t, int k) {
        if (t == -1) return {-1, -1};
        push(t);
        if (getSize(tree[t].l) >= k) {
            auto res = split(tree[t].l, k);
            tree[t].l = res.second; pull(t);
            return {res.first, t};
        } else {
            auto res = split(tree[t].r, k -
                getSize(tree[t].l) - 1);
            tree[t].r = res.first; pull(t);
            return {t, res.second};
        }
    }
    int merge(int a, int b) {
        if (a == -1) return b;
        if (b == -1) return a;
        if (tree[a].priority > tree[b].
            priority) {
            push(a); tree[a].r = merge(tree[
                a].r, b); pull(a);
            return a;
        } else {
            push(b); tree[b].l = merge(a, tree
                [b].l); pull(b);
            return b;
        }
    }
    int insert(int root, int pos, int val)
        {
        int nn = newNode(val);
        auto [L, R] = split(root, pos);
        return merge(merge(L, nn), R);
    }
    int erase(int root, int pos) {
        auto [L, R] = split(root, pos);
        auto [p, q] = split(R, 1);
        return merge(L, q);
    }
    void show(int t) {
        if (t == -1) return;
        push(t); show(tree[t].l);
        cout << tree[t].val << " "; show(
            tree[t].r);
    }
};

```

4.10 Segment Tree 2D (1 based)

```

// USAGE
ST2D sg(NX, NY); // dimensions 1..NX,
1..NY
sg.upd (1, 1, NX, x, y, v);

```

```

sg.query(1, 1, NX, x1, x2, y1, y2);
struct ST2D {
    #define lc (nd << 1)
    #define rc ((nd << 1) | 1)
    #define lcy (ndy << 1)
    #define rcy ((ndy << 1) | 1)
    int NX, NY;
    vector<vector<long long>> t;
    ST2D(int NX, int NY) : NX(NX), NY(NY),
        t(4 * NX, vector<long long>(4 * NY, 0)) {}
    inline long long combine(long long a,
        long long b) { return max(a, b); }
    inline void pull_y(int nd, int ndy) {
        t[nd][ndy] = combine(t[nd][lcy], t[nd][rcy]);
    }
    inline void pull_x(int nd, int ndy) {
        t[nd][ndy] = combine(t[lc][ndy], t[rc][ndy]);
    }
    // -----
    // Inner seg tree on y-axis
    // -----
    void upd_y(int nd, int ndy, int by,
        int ey, int j, long long v, bool
        leaf_x) {
        if (by == ey) {
            if (leaf_x) t[nd][ndy] = v;
            else t[nd][ndy] = combine(t[lc][ndy], t[rc][ndy]);
            return;
        }
        int mid = (by + ey) >> 1;
        if (j <= mid) upd_y(nd, lcy, by, mid,
            j, v, leaf_x);
        else upd_y(nd, rcy, mid+1,
            ey, j, v, leaf_x);
        pull_y(nd, ndy);
    }
    long long query_y(int nd, int ndy, int
        by, int ey, int y1, int y2) {
        if (y1 > ey || by > y2) return 0;
        if (y1 <= by && ey <= y2) return t[nd][ndy];
        int mid = (by + ey) >> 1;
        return combine(query_y(nd, lcy, by,
            mid, y1, y2),
            query_y(nd, rcy, mid
                +1, ey, y1, y2));
    }
    // -----
    // Outer seg tree on x-axis
    // -----
    void upd(int nd, int bx, int ex, int i
        , int j, long long v) {
        if (bx == ex) {
            upd_y(nd, 1, 1, NY, j, v, true);
            return;
        }
        int mid = (bx + ex) >> 1;
        if (i <= mid) upd(lc, bx, mid, i, j,
            v);
        else upd(rc, mid+1, ex, i,
            j, v);
        upd_y(nd, 1, 1, NY, j, v, false);
        // pull x into inner nodes
    }
    long long query(int nd, int bx, int ex
        , int x1, int x2, int y1, int y2) {
        if (x1 > ex || bx > x2) return 0;
        if (x1 <= bx && ex <= x2) return
            query_y(nd, 1, 1, NY, y1, y2);
        int mid = (bx + ex) >> 1;
        return combine(query(lc, bx, mid, x1
            , x2, y1, y2),
            query(rc, mid+1, ex,
                x1, x2, y1, y2));
    }
};

```

4.11 Segment Tree 2D (Index compression)

```

// USAGE
ST2D seg(n);
// Phase 1: register all (x, y) points

```

```

that will ever be updated
for (auto& [x, y, v] : updates)
    seg.collect(1, 1, n, x, y);
// Phase 2: compress and allocate inner
// trees
seg.build(1, 1, n);
// Updates and queries
seg.upd(1, 1, n, x, y, v);
seg.query(1, 1, n, x1, x2, y1, y2);
//
// =====
void compress(vector<int>& a) {
    sort(all(a));
    a.erase(unique(all(a)), a.end());
}
int ind(vi& a, int val) {
    return lower_bound(all(a), val) - a.
        begin();
}
struct ST2D {
    #define lc (nd << 1)
    #define rc ((nd << 1) | 1)
    int n;
    vector<vi> ys; // compressed y-
        coords per x-node
    vector<vi> t; // inner seg tree
        per x-node
    ST2D(int n) : n(n), ys(4 * n), t(4 * n) {}
    void collect(int nd, int b, int e, int
        x, int y) {
        ys[nd].pb(y);
        if (b == e) return;
        int mid = (b + e) >> 1;
        if (x <= mid) collect(lc, b, mid, x,
            y);
        else collect(rc, mid + 1, e,
            x, y);
    }
    void build(int nd, int b, int e) {
        compress(ys[nd]);
        t[nd].assign(ys[nd].sz() * 2, 0);
        if (b == e) return;
        int mid = (b + e) >> 1;
        build(lc, b, mid);
        build(rc, mid + 1, e);
    }
    inline long long combine(long long a,
        long long b) { return max(a, b); }
    void upd_y(int nd, int y, long long v) {
        int m = ys[nd].sz();
        int i = ind(ys[nd], y) + m;
        t[nd][i] = combine(t[nd][i], v);
        for (i >>= 1; i >= 1; i >>= 1)
            t[nd][i] = combine(t[nd][i*2], t[nd][i*2+1]);
    }
    long long query_y(int nd, int y1, int
        y2) {
        if (y1 > y2) return 0;
        int m = ys[nd].sz();
        int l = ind(ys[nd], y1) + m;
        int r = (int)upper_bound(all(ys[nd]),
            y2) - ys[nd].begin() - 1 + m;
        ll mx = 0;
        while (l <= r) {
            if (l%2 == 1) mx = combine(mx, t[nd][l++]);
            if (r%2 == 0) mx = combine(mx, t[nd][r--]);
            l >>= 1; r >>= 1;
        }
        return mx;
    }
    void upd(int nd, int b, int e, int x,
        int y, long long v) {
        upd_y(nd, y, v);
        if (b == e) return;
        int mid = (b + e) >> 1;
        if (x <= mid) upd(lc, b, mid, x, y,
            v);
        else upd(rc, mid + 1, e, x,
            y, v);
    }
    long long query(int nd, int b, int e,
        int x1, int x2, int y1, int y2) {

```

```

        if (x1 > x2 || y1 > y2) return 0;
        // empty range (e.g. a[i].f == 1)
        if (x1 > e || b > x2) return 0;
        if (x1 <= b && e <= x2) return
            query_y(nd, y1, y2);
        int mid = (b + e) >> 1;
        return combine(query(lc, b, mid, x1,
            x2, y1, y2),
            query(rc, mid+1, e,
                x1, x2, y1, y2));
    }
};

```

4.12 DSU

```

struct DSU {
    vector<int> par, sz;
    DSU(){}
    DSU(int n){
        init(n);
    }
    void init(int n){
        par.resize(n);
        iota(par.begin(), par.end(), 0);
        sz.assign(n, 1);
    }
    int find(int v){
        while(v != par[v]){
            v = par[v] = par[par[v]];
        }
        return v;
    }
    bool merge(int a, int b){
        a = find(a);
        b = find(b);
        if (a == b) return false;
        if (sz[a] < sz[b]) swap(a, b);
        par[b] = a;
        sz[a] += sz[b];
        return true;
    }
};

```

4.13 SQRT Decomposition

```

struct SqrtDecomp {
    int n, sz;
    vector<ll> a;
    vector<map<int, int>> b;
    SqrtDecomp(int n, vi& arr) : n(n) {
        sz = sqrt(n) + 1;
        /*...*/
    }
    void upd(int i, ll v) {
        /*...*/
    }
    ll combine(ll s, ll elem, int v) {
        /*...*/
    }
    ll combineBlock(ll s, map<int, int>&
        blk, int v) {
        /*...*/
    }
    ll query(int l, int r, int v) {
        ll s = 0;
        r--;
        if (l > r) {
            return 0;
        }
        int bl = l/sz, br = r/sz;
        if (bl == br) {
            for (int i = l; i <= r; i++)
                s = combine(s, a[i], v);
        } else {
            for (int i = l; i < (bl+1)*
                sz; i++) s = combine(s, a[i], v);
            for (int i = bl+1; i < br; i++)
                s = combineBlock(s, b[i], v);
            for (int i = br*sz; i <= r; i++)
                s = combine(s, a[i], v);
        }
        return s;
    }
};

```

4.14 MOs Algorithm

```

struct MO{
    int n,q,B,res; struct query{ int l,r,
        id; };
    vector<query> Q; vector<int> ans,cnt;
    MO(int n, int q): n(n), q(q){
        res=0;
        B=sqrt(n)+1;
        ans.resize(q+1);
        cnt.resize(n+10);
    }
    void add_query(int l, int r, int id){
        Q.push_back({l,r,id});
    }
    void add_left(int id,const vector<int>
        &a){}
    void add_right(int id,const vector<
        int>&a){}
    void rem_left(int id,const vector<int>
        &a){}
    void rem_right(int id,const vector<
        int>&a){}
    void get_ans(const vector<int>&a){
        sort(Q.begin(),Q.end(),[&](auto& x
        , auto& y){
            if(x.l/B == y.l/B){
                return ((x.l/B)&1) ? x.r>y.r
                : x.r<y.r;
            }
            return x.l/B < y.l/B;
        });
        int L=1,R=0;
        for(auto &[l,r,id]: Q){
            while(L>l) add_left(--L,a);
            while(R<r) add_right(++R,a);
            while(L<l) rem_left(L++,a);
            while(R>r) rem_right(R--,a);
            ans[id]=res;
        }
    }
};

```

4.15 MOs with updates

```

struct mos_update {
    ll a[100005];
    ll ans;
    mos_update() { /* ... */ }
    void add(int ind, int end, int L, int
        R) { /* ... */ }
    void del(int ind, int end, int L, int
        R) { /* ... */ }
    void apply(int pos, int val, int L,
        int R) {
        if (L <= pos && pos <= R) {
            del(pos, -1, L, R); a[pos] = val;
            add(pos, -1, L, R);
        } else {
            a[pos] = val;
        }
    }
    vector<ll> mo(vector<vector<int>>& Q,
        vector<vector<int>>& U, int n) {
        int L = 1, R = 0, T = 0;
        ans = 0;
        int blk = max(1, (int)pow(Q.size(),
            0.666));
        vector<int> s(Q.size());
        vector<ll> res(Q.size());
        iota(s.begin(), s.end(), 0);
        sort(s.begin(), s.end(), [&](int i,
            int j) {
            int li = Q[i][0] / blk, lj = Q[j]
            [0] / blk;
            if (li != lj) return li < lj;
            int ri = Q[i][1] / blk, rj = Q[j]
            [1] / blk;
            if (ri != rj) return (li & 1) ? ri
            < rj : ri > rj;
            return (ri & 1) ? Q[i][2] < Q[j]
            [2] : Q[i][2] > Q[j][2];
        });
        for (int qi : s) {
            int ql = Q[qi][0], qr = Q[qi][1],
            qt = Q[qi][2];
            while (T < qt) { apply(U[T][0], U[
                T][2], L, R); T++; }
            while (T > qt) { --T; apply(U[T]
                [0], U[T][1], L, R); }
        }
    }
};

```

```

        while (L > ql) { --L; add(L, 0, L,
            R); }
        while (R < qr) { ++R; add(R, 1, L,
            R); }
        while (L < ql) { del(L, 0, L, R);
            L++; }
        while (R > qr) { del(R, 1, L, R);
            R--; }
        res[qi] = ans;
    }
    return res;
}
};

```

4.16 Treap

```

struct ImplicitTreap {
    struct Node { int l, r, sz, sum, val,
        priority, lazy; };
    vector<Node> tree;
    vector<int> rnd;
    ImplicitTreap(int max_nodes = 0) {
        rnd.seed(chrono::steady_clock::now()
            .time_since_epoch().count());
        if (max_nodes > 0) tree.reserve(
            max_nodes);
    }
    int getSize(int t) { return t == -1 ?
        0 : tree[t].sz; }
    int getSum(int t) { return t == -1 ? 0
        : tree[t].sum; }
    void applyLazy(int t, int add) {
        if (t == -1) return;
        tree[t].val += add; tree[t].sum +=
        tree[t].sz * add; tree[t].lazy +=
        add;
    }
    void pull(int t) {
        if (t == -1) return;
        tree[t].sz = 1 + getSize(tree[t].l)
        + getSize(tree[t].r);
        tree[t].sum = tree[t].val + getSum(
            tree[t].l) + getSum(tree[t].r);
    }
    void push(int t) {
        if (t == -1 || tree[t].lazy == 0)
            return;
        applyLazy(tree[t].l, tree[t].lazy);
        applyLazy(tree[t].r, tree[t].lazy);
        tree[t].lazy = 0;
    }
    int newNode(int val) {
        tree.push_back({-1, -1, 1, val, val,
            (int)rnd(), 0});
        return tree.size() - 1;
    }
    pair<int, int> split(int t, int k) {
        if (t == -1) return {-1, -1};
        push(t);
        if (getSize(tree[t].l) >= k) {
            auto res = split(tree[t].l, k);
            tree[t].l = res.second; pull(t);
            return {res.first, t};
        } else {
            auto res = split(tree[t].r, k -
                getSize(tree[t].l) - 1);
            tree[t].r = res.first; pull(t);
            return {t, res.second};
        }
    }
    int merge(int a, int b) {
        if (a == -1) return b;
        if (b == -1) return a;
        if (tree[a].priority > tree[b].
            priority) {
            push(a); tree[a].r = merge(tree[a]
                .r, b); pull(a);
            return a;
        } else {
            push(b); tree[b].l = merge(a, tree
                [b].l); pull(b);
            return b;
        }
    }
    int insert(int root, int pos, int val) {
        int nn = newNode(val);
    }
};

```

```

        auto [L, R] = split(root, pos);
        return merge(merge(L, nn), R);
    }
    int erase(int root, int pos) {
        auto [L, R] = split(root, pos);
        auto [p, q] = split(R, 1);
        return merge(L, q);
    }
    void show(int t) {
        if (t == -1) return;
        push(t); show(tree[t].l);
        cout << tree[t].val << " "; show(
            tree[t].r);
    }
};

```

4.17 Wavelet Tree

```

struct wavelet_tree {
    int lo, hi;
    wavelet_tree *l, *r;
    vector<int> b, c;
    wavelet_tree() {
        lo = hi = 0;
        l = r = NULL;
    }
    void init(int *from, int *to, int x,
        int y) {
        lo = x, hi = y;
        if (from >= to) return;
        int mid = (lo + hi) >> 1;
        auto f = [mid](int x) { return x
            <= mid; };
        b.reserve(to - from + 1);
        c.reserve(to - from + 1);
        b.push_back(0);
        c.push_back(0);
        for (auto it = from; it != to;
            ++it) {
            b.push_back(b.back() + f(*it));
            c.push_back(c.back() + (*it));
        }
        if (hi == lo) return;
        auto pivot = stable_partition(
            from, to, f);
        l = new wavelet_tree(); l->init(
            from, pivot, lo, mid);
        r = new wavelet_tree(); r->init(
            pivot, to, mid + 1, hi);
    }
    int kth(int l, int r, int k) {
        if (l > r) return 0;
        if (lo == hi) return lo;
        int inLeft = b[r] - b[l - 1], lb
            = b[l - 1], rb = b[r];
        if (k <= inLeft) return this->l
            ->kth(lb + 1, rb, k);
        return this->r->kth(l - lb, r -
            rb, k - inLeft);
    }
    int LTE(int l, int r, int k) {
        if (l > r || k < lo) return 0;
        if (hi <= k) return r - l + 1;
        int lb = b[l - 1], rb = b[r];
        return this->l->LTE(lb + 1, rb,
            k) + this->r->LTE(l - lb, r - rb, k);
    }
    int count(int l, int r, int k) {
        if (l > r || k < lo || k > hi)
            return 0;
        if (lo == hi) return r - l + 1;
        int lb = b[l - 1], rb = b[r];
        mid = (lo + hi) >> 1;
        if (k <= mid) return this->l->
            count(lb + 1, rb, k);
        return this->r->count(l - lb, r
            - rb, k);
    }
    int sum(int l, int r, int k) {
        if (l > r || k < lo) return 0;
        if (hi <= k) return c[r] - c[l -
            1];
        int lb = b[l - 1], rb = b[r];
        return this->l->sum(lb + 1, rb,
    }
};

```

```

    k) + this->r->sum(l - lb, r - rb, k
    );
}
~wavelet_tree() { delete l; delete r
; }
};
wavelet_tree t;

```

4.18 Sparse Table (RMQ)

```

template<typename T>
struct SparseTable {
    const int N = 1e5 + 9;
    vector<vector<T>>> t;
    vector<T> a;
    T combine(T x, T y) { return min(x, y)
    ; }
    void build(int n) {
        t.assign(n + 1, vector<T>(18));
        for(int i = 1; i <= n; ++i) t[i][0]
        = a[i];
        for(int k = 1; k < 18; ++k)
            for(int i = 1; i + (1 << k) - 1 <=
            n; ++i)
                t[i][k] = combine(t[i][k-1], t[i
                + (1 << (k-1))][k-1]);
    }
    T query(int l, int r) {
        int k = 31 - __builtin_clz(r - l +
        1);
        return combine(t[l][k], t[r - (1 <<
        k) + 1][k]);
    }
};

```

4.19 Prefix Sum 2D

```

template<typename T>
struct PrefixSum2D {
    vector<vector<T>>> p;
    int rows, cols;
    T combine(T x, T y) { return x + y; }
    void build(vector<vector<T>>& a) {
        rows = a.size(); cols = a[0].size();
        p.assign(rows + 1, vector<T>(cols +
        1, 0));
        for(int i = 1; i <= rows; ++i)
            for(int j = 1; j <= cols; ++j)
                p[i][j] = a[i-1][j-1] + p[i-1][j
                ] + p[i][j-1] - p[i-1][j-1];
    }
    // query sum over [r1,c1]..[r2,c2] (1-
    indexed)
    T query(int r1, int c1, int r2, int c2
    ) {
        return p[r2][c2] - p[r1-1][c2] - p[
        r2][c1-1] + p[r1-1][c1-1];
    }
};

```

4.20 Prefix Sum 3D

```

template<typename T>
struct PrefixSum3D {
    vector<vector<vector<T>>>> p;
    int X, Y, Z;
    T combine(T x, T y) { return x + y; }
    void build(vector<vector<vector<T>>>&
    a) {
        X = a.size(); Y = a[0].size(); Z = a
        [0][0].size();
        p.assign(X+1, vector<vector<T>>>(Y+1,
        vector<T>(Z+1, 0)));
        for(int i = 1; i <= X; ++i)
            for(int j = 1; j <= Y; ++j)
                for(int k = 1; k <= Z; ++k)
                    p[i][j][k] = a[i-1][j-1][k-1]
                    + p[i-1][j][k] + p[i][j-1][k
                    ] + p[i][j][k-1]
                    - p[i-1][j-1][k] - p[i-1][j
                    ][k-1] - p[i][j-1][k-1]
                    + p[i-1][j-1][k-1];
    }
    // query sum over [x1,y1,z1]..[x2,y2,
    z2] (1-indexed)
    T query(int x1, int y1, int z1, int x2
    , int y2, int z2) {

```

```

        return p[x2][y2][z2]
        - p[x1-1][y2][z2] - p[x2][y1-1][z2
        ] - p[x2][y2][z1-1]
        + p[x1-1][y1-1][z2] + p[x1-1][y2][
        z1-1] + p[x2][y1-1][z1-1]
        - p[x1-1][y1-1][z1-1];
    }
};

```

5 Graph

5.1 Dijkstra

```

struct Node {
    int u;
    ll dist;
    bool operator<(const Node& v) const {
        return dist > v.dist; }
};
vector<ll> dijkstra(int start, vector<
VPI&& adj) {
    vector<ll> dis(adj.size(), INFL);
    priority_queue<Node> q;
    dis[start] = 0;
    q.push({start, 0});
    while (!q.empty()) {
        Node node = q.top();
        int u = node.u;
        q.pop();
        if (dis[u] < node.dist) continue;
        for (auto e : adj[u]) {
            if (node.dist + e.ss < dis[e.ff])
            {
                dis[e.ff] = node.dist + e.ss;
                q.push({e.ff, dis[e.ff]});
            }
        }
    }
    return dis;
}

```

5.2 Bellmanford

```

const ll inf = LLONG_MAX;
struct Ed { int a, b, w; ll s() { return
a < b ? a : -a; }};
struct Node { ll dist = inf; int prev =
-1; };
void bellmanFord(vector<Node>& nodes,
vector<Ed>& eds, int s) {
    nodes[s].dist = 0;
    sort(all(eds), [](Ed a, Ed b) { return
a.s() < b.s(); });
    int lim = nodes.size() / 2 + 2; //
/3+100 with shuffled vertices
    for(int i = 0; i < lim; i++) for (Ed
ed : eds) {
        Node cur = nodes[ed.a], &dest =
nodes[ed.b];
        if (abs(cur.dist) == inf) continue;
        ll d = cur.dist + ed.w;
        if (d < dest.dist) {
            dest.prev = ed.a;
            dest.dist = (i < lim-1 ? d : -inf)
            ;
        }
    }
    for(int i = 0; i < lim; i++) for (Ed e
: eds) {
        if (nodes[e.a].dist == -inf)
            nodes[e.b].dist = -inf;
    }
}

```

5.3 BCC, Bridge, AP

```

// g[u] = {{neighbor, edge_id}, ...}
vector<vector<pair<int,int>>> g;
vector<int> tin, st, bridges;
vector<vector<int>> bccs;
int timer;
int dfs(int u, int pe) {
    int me = tin[u] = ++timer, top = me,
    ch = 0;

```

```

    for (auto [v, id] : g[u]) if (id != pe
    ) {
        if (tin[v]) {
            top = min(top, tin[v]);
            if (tin[v] < me) st.push_back(id);
        } else {
            ch++;
            int si = st.size();
            int up = dfs(v, id);
            top = min(top, up);
            if (up >= me) {
                if (pe != -1) is_art[u] = true;
                if (up == me) {
                    st.push_back(id);
                    bccs.push_back({st.begin() +
                    si, st.end()});
                    st.resize(si);
                } else {
                    bridges.push_back(id);
                }
            } else {
                st.push_back(id);
            }
        }
    }
    if (pe == -1 && ch > 1) is_art[u] =
    true;
    return top;
}
void get_bcc(int n) {
    tin.assign(n + 1, 0); is_art.assign(n
    + 1, false);
    st.clear(); bccs.clear(); bridges.
    clear(); timer = 0;
    for (int i = 1; i <= n; i++) if (!tin[
    i]) dfs(i, -1);
}

```

5.4 2CC

```

struct TECC{
    using Edge=array<int,2>;
    int n,cnt_comp;
    vector<vector<int>>> G,T; vector<bool>
    vis;
    vector<int> tin,low,comp; vector<Edge>
    bridges;
    void low_dfs(int u, int par, int& cur)
    {
        vis[u]=true; tin[u]=cur++; low[u]=n
        +1;
        bool usedPar=false;
        for(auto v: G[u]){
            if(vis[v]){
                if(v!=par || usedPar){
                    low[u]=min(low[u],tin[v]);
                }else{
                    usedPar=true;
                }
                continue;
            }
            low_dfs(v,u,cur);
            low[u]=min(low[u],low[v]);
        }
    }
    void build_dfs(int u, int cid){
        comp[u]=cid;
        for(auto v: G[u]){
            if(comp[v]!=-1) continue;
            if(tin[u]<low[v]){
                bridges.push_back({u,v});
                cnt_comp++;
                build_dfs(v,cnt_comp);
            }else{
                build_dfs(v,cid);
            }
        }
    }
    TECC(int _n,const vector<vector<int>>&
    g){
        n=_n; G=g; cnt_comp=1;
        tin.resize(n+1);
        low.resize(n+1);
        vis.assign(n+1,false);
        comp.assign(n+1,-1);
        int cur=1;
        for(int i=1;i<=n;i++){

```

```

    if(vis[i]) continue;
    low_dfs(i,-1,cur);
    build_dfs(i,cnt_comp);
    cnt_comp++;
}
}
void build_tree(){
    T.assign(cnt_comp,{});
    for(auto &e: bridges){
        int u=comp[e[0]],v=comp[e[1]];
        T[u].push_back(v);
        T[v].push_back(u);
    }
}
};

```

5.5 LCA

```

struct LCA {
    int step = 0, k = 0;
    vector<int> vis_start, vis_end;
    vector<vector<int>> st;
    vector<vector<int>>& tree;
    LCA(vector<vector<int>>& tree) : tree(
        tree) { init(tree.size() - 1); }
    void build_st(int u, int p) {
        st[u][0] = p;
        for (int i = 1; i <= k; ++i) st[u][i]
            = st[st[u][i - 1]][i - 1];
    }
    void dfs(int u, int p) {
        vis_start[u] = step++;
        build_st(u, p);
        for (auto v : tree[u]) {
            if (v != p) dfs(v, u);
        }
        vis_end[u] = step++;
    }
    bool is_ancestor(int a, int b) {
        return vis_start[a] < vis_start[b]
            && vis_end[a] > vis_end[b];
    }
    int get(int a, int b) {
        if (a == b) return a;
        if (is_ancestor(a, b)) return a;
        if (is_ancestor(b, a)) return b;
        for (int i = k; i >= 0; --i) {
            if (!is_ancestor(st[a][i], b)) a =
                st[a][i];
        }
        return st[a][0];
    }
    void init(int n) {
        vis_start.assign(n + 1, 0);
        vis_end.assign(n + 1, 0);
        k = __lg(2 * n - 1);
        st.assign(n + 1, vector<int>(k + 1));
        dfs(1, 1);
    }
    // Extra
    int get_ancestor(int u, int n) {
        for (int i = 0; n > 0; ++i) {
            if (n & (1 << i)) {
                u = st[u][i];
                n -= (1 << i);
            }
        }
        return u;
    }
};

```

5.6 SCC

```

vector<int> val, comp, z;
int Time, ncomps;
vector<vector<int>> sccs;
int dfs(int j, vector<vector<int>>& g) {
    int low = val[j] = ++Time, x; z.
        push_back(j);
    for (int e : g[j]) if (comp[e] < 0)
        low = min(low, val[e] ? dfs(e,
            g));
    if (low == val[j]) {
        sccs.push_back({});
        do {
            x = z.back(); z.pop_back();

```

```

        comp[x] = ncomps;
        sccs.back().push_back(x);
    } while (x != j);
    ncomps++;
}
return val[j] = low;
}
void get_sccs(vector<vector<int>>& g) {
    int n = g.size();
    val.assign(n, 0); comp.assign(n, -1);
    ;
    z.clear(); sccs.clear();
    Time = ncomps = 0;
    for (int i = 0; i < n; i++) if (comp
        [i] < 0) dfs(i, g);
}

```

5.7 Kosaraju

```

class Kosaraju{
public:
    int n;
    vector<vector<int32_t>> g, rg;
    vector<int32_t> comp, order, sz; vector<
        bool> vis;
    void dfs1(int u) {
        vis[u] = true;
        for (int v : g[u]) {
            if (!vis[v]) dfs1(v);
        }
        order.push_back(u);
    }
    void dfs2(int u, int c) {
        comp[u] = c; sz[comp[u]]++;
        for (auto& v : rg[u]) {
            if (comp[v] == -1) dfs2(v, c);
        }
    }
    Kosaraju(int n) : n(n) {
        g.assign(n + 1, {});
        rg.assign(n + 1, {});
        sz.assign(n + 1, 0);
    }
    inline void add_edge(int u, int v) {
        g[u].push_back(v);
        rg[v].push_back(u);
    }
    int build() {
        vis.assign(n + 1, false);
        order.clear();
        for (int i = 1; i <= n; i++) {
            if (!vis[i]) dfs1(i);
        }
        comp.assign(n + 1, -1);
        reverse(order.begin(), order.end());
        int scc = 0;
        for (int u : order) {
            if (comp[u] == -1) {
                dfs2(u, ++scc);
            }
        }
        return scc;
    }
    vector<vector<int32_t>> build_dag(int
        scc) {
        vector<vector<int32_t>> dag(scc + 1)
            ;
        for (int u = 1; u <= n; u++) {
            for (int v : g[u]) {
                if (comp[u] != comp[v]){
                    dag[comp[u]].push_back(comp[v]
                        );
                }
            }
        }
        return dag;
    }
};

```

5.8 HLD

```

vector<int> heavy, chain, label, par, sz
    , dep;
int timer;
void dfsz(int u, int p, vector<vector<
    int>>& tree) {
    sz[u] = 1;

```

```

    par[u] = p;
    dep[u] = dep[p] + 1;
    int mx = -1;
    for (int v : tree[u]) {
        if (v == p) continue;
        dfsz(v, u, tree);
        sz[u] += sz[v];
        if (sz[v] > mx) mx = sz[v], heavy[u]
            = v;
    }
}
void dfshld(int u, int p, vector<vector<
    int>>& tree, int head) {
    label[u] = timer++;
    chain[u] = head;
    if (heavy[u] != -1) dfshld(heavy[u], u
        , tree, head);
    for (int v : tree[u]) {
        if (v == p || v == heavy[u])
            continue;
        dfshld(v, u, tree, v);
    }
}
void inithld(vector<vector<int>>& tree)
    {
    int n = tree.size();
    heavy = vector<int>(n, -1);
    chain = vector<int>(n);
    label = vector<int>(n);
    sz = vector<int>(n);
    par = vector<int>(n);
    dep = vector<int>(n);
    timer = 1;
    dfsz(1, 0, tree);
    dfshld(1, 0, tree, 1);
}
// For path update, you can use the
// query function itself, change
// query with update
ll query(int u, int v, /*, fenwick/
    segtree/anything& ds*/) {
    ll ret = 0; // init value
    for (; chain[u] != chain[v]; v = par[
        chain[v]]) {
        if (dep[chain[u]] > dep[chain[v]])
            swap(u, v);
        // ret += ds.query(label[chain[v]],
            label[u]);
    }
    if (dep[u] > dep[v]) swap(u, v);
    // lca is u
    // For EDGES, do label[u]+1
    // ret += ds.query(label[u], label[v]);
    return ret;
}
ll querySubtree(int u, /*, fenwick/
    segtree/anything& ds */) {
    // For EDGES, do label[u]+1
    return // ds.query(label[u], label[u]
        ]+sz[u]-1)
}
// Point Updating u : ds.update(label[u]
    ,...)

```

5.9 Centroid Decomposition

```

struct CD {
    vector<vector<int>>& tree;
    vector<int> par, sz;
    vector<bool> dead;
    CD(vector<vector<int>>& tree) : tree(
        tree) {
        par.resize(tree.size());
        sz.resize(tree.size());
        dead.resize(tree.size());
        build(1, 0);
    }
    void build(int u, int p) {
        dfs(u, p);
        u = dfs(u, p, sz[u]);
        dead[u] = true;
        par[u] = p;
        // u is centroid, process here...
        for (int v : tree[u]) {
            if (!dead[v]) build(v, u);
        }
    }
}

```

```

}
void dfs(int u, int p) {
    sz[u] = 1;
    for (int v : tree[u]) {
        if (v == p || dead[v]) continue;
        dfs(v, u);
        sz[u] += sz[v];
    }
}
int dfs(int u, int p, int n) {
    for (int v : tree[u]) {
        if (v == p || dead[v]) continue;
        if (sz[v] > n / 2) return dfs(v, u, n);
    }
    return u;
}
int operator[](int i) { return par[i]; }
};

```

5.10 Arpa's Trick

```

const int N = 1e5 + 9;
vector<int> g[N];
int ans[N], col[N], sz[N], cnt[N];
bool big[N];
void dfs(int u, int p) {
    sz[u] = 1;
    for (auto v : g[u]) {
        if (v == p) continue;
        dfs(v, u);
        sz[u] += sz[v];
    }
}
void add(int u, int p, int x) {
    cnt[col[u]] += x;
    for (auto v : g[u]) {
        if (v == p || big[v] == 1) continue;
        add(v, u, x);
    }
}
void dsu(int u, int p, bool keep) {
    int bigchild = -1, mx = -1;
    for (auto v : g[u]) {
        if (v == p) continue;
        if (sz[v] > mx) mx = sz[v], bigchild = v;
    }
    for (auto v : g[u]) {
        if (v == p || v == bigchild) continue;
        dsu(v, u, 0);
    }
    if (bigchild != -1) dsu(bigchild, u, 1), big[bigchild] = 1;
    add(u, p, 1);
    ans[u] = cnt[u];
    if (bigchild != -1) big[bigchild] = 0;
    if (keep == 0) add(u, p, -1);
}

```

6 Flow and Matching

6.1 Min Cost Max Flow

```

const ll INF = numeric_limits<ll>::max() / 4;
struct MCMF {
    struct edge {
        int from, to, rev;
        ll cap, cost, flow;
    };
    int N;
    vector<vector<edge>> ed;
    vi seen;
    vector<ll> dist, pi;
    vector<edge*> par;
    MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N) {}
    void addEdge(int from, int to, ll cap, ll cost) {
        if (from == to) return;

```

```

        ed[from].push_back(edge{from, to, sz(ed[to]), cap, cost, 0});
        ed[to].push_back(edge{to, from, sz(ed[from]) - 1, 0, -cost, 0});
    }
    void path(int s) {
        fill(all(seen), 0);
        fill(all(dist), INF);
        dist[s] = 0;
        ll di;
        __gnu_pbds::priority_queue<pair<ll, int>> q;
        vector<decltype(q)::point_iterator> its(N);
        q.push({0, s});
        while (!q.empty()) {
            s = q.top().second;
            q.pop();
            seen[s] = 1;
            di = dist[s] + pi[s];
            for (edge& e : ed[s])
                if (!seen[e.to]) {
                    ll val = di - pi[e.to] + e.cost;
                    if (e.cap - e.flow > 0 && val < dist[e.to]) {
                        dist[e.to] = val;
                        par[e.to] = &e;
                        if (its[e.to] == q.end()) its[e.to] = q.push({-dist[e.to], e.to});
                        else q.modify(its[e.to], {-dist[e.to], e.to});
                    }
                }
        }
        rep(i, 0, N) pi[i] = min(pi[i] + dist[i], INF);
    }
    pair<ll, ll> maxflow(int s, int t) {
        ll totflow = 0, totcost = 0;
        while (path(s), seen[t]) {
            ll fl = INF;
            for (edge* x = par[t]; x; x = par[x->from])
                fl = min(fl, x->cap - x->flow);
            totflow += fl;
            for (edge* x = par[t]; x; x = par[x->from]) {
                x->flow += fl;
                ed[x->to][x->rev].flow -= fl;
            }
        }
        rep(i, 0, N) for (edge& e : ed[i]) totcost += e.cost * e.flow;
        return {totflow, totcost / 2};
    }
    // If some costs can be negative, call this before maxflow:
    void setpi(int s) { // (otherwise, leave this out)
        fill(all(pi), INF);
        pi[s] = 0;
        int it = N, ch = 1;
        ll v;
        while (ch-- && it--)
            rep(i, 0, N) if (pi[i] != INF) for (edge& e : ed[i])
                if (e.cap) if ((v = pi[i] + e.cost) < pi[e.to]) pi[e.to] = v, ch = 1;
        assert(it >= 0); // negative cost cycle
    }
};

```

6.2 Dinic

```

const int64_t inf = 1LL << 60;
struct Dinic {

```

```

    struct Edge {
        int to, rev;
        int64_t flow, w;
        int id;
    };
    int n, s, t, mxid;
    vector<int> d, done;
    vector<int64_t> flow_through;
    vector<vector<Edge>> g;
    Dinic() {}
    Dinic(int _n) {
        n = _n + 10;
        mxid = 0;
        g.resize(n);
    }
    void add_edge(int u, int v, int64_t w, int id = -1) {
        Edge a = {v, (int)g[v].size(), 0, w, id};
        Edge b = {u, (int)g[u].size(), 0, 0, -2}; // cap(b) = w for bidirectional
        g[u].emplace_back(a);
        g[v].emplace_back(b);
        if (id >= 0) mxid = max(mxid, id);
    }
    bool bfs() {
        d.assign(n, -1);
        d[s] = 0;
        queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            for (auto &e : g[u]) {
                int v = e.to;
                if (d[v] == -1 && e.flow < e.w) {
                    d[v] = d[u] + 1;
                    q.push(v);
                }
            }
        }
        return d[t] != -1;
    }
    int64_t dfs(int u, int64_t flow) {
        if (u == t) return flow;
        for (int &i = done[u]; i < (int)g[u].size(); i++) {
            Edge &e = g[u][i];
            if (e.w <= e.flow) continue;
            int v = e.to;
            if (d[v] == d[u] + 1) {
                int64_t nw = dfs(v, min(flow, e.w - e.flow));
                if (nw > 0) {
                    e.flow += nw;
                    g[v][e.rev].flow -= nw;
                    return nw;
                }
            }
        }
        return 0;
    }
    int64_t max_flow(int _s, int _t) {
        s = _s;
        t = _t;
        int64_t flow = 0;
        while (bfs()) {
            done.assign(n, 0);
            while (int64_t nw = dfs(s, inf)) flow += nw;
        }
        flow_through.assign(mxid+10, int64_t(0));
        for (int i = 0; i < n; i++) {
            for (auto &e : g[i]) {
                if (e.id >= 0) {
                    flow_through[e.id] = e.flow;
                }
            }
        }
        return flow;
    }
};

```

6.3 HopcroftKarp

```
int hopcroftKarp(vector<vi>& g, vi& r) {
    int n = sz(g), res = 0;
    vi l(n, -1), q(n), d(n);
    auto dfs = [&](auto f, int u) -> bool {
        int t = exchange(d[u], 0) + 1;
        for (int v : g[u])
            if (r[v] == -1 || (d[r[v]] == t && f(f, r[v])))
                return l[u] = v, r[v] = u, 1;
        return 0;
    };
    for (int t = 0, f = 0;; t = f = 0, d.assign(n, 0)) {
        rep(i, 0, n) if (l[i] == -1) q[t++] = i, d[i] = 1;
        rep(i, 0, t) for (int v : g[q[i]]) {
            if (r[v] == -1)
                f = 1;
            else if (!d[r[v]])
                d[r[v]] = d[q[i]] + 1, q[t++] = r[v];
        }
        if (!f) return res;
        rep(i, 0, n) if (l[i] == -1) res += dfs(dfs, i);
    }
}
// --- Shortened usage ---
int main() {
    int n, m, k; cin >> n >> m >> k;
    vi r(n + m, -1); vector<vi> g(n + m);
    rep(i, 0, k) {
        int u, v; cin >> u >> v;
        g[u-1].push_back(v + n - 1); // Map right side to [n, n+m]
    }
    int matchingSize = hopcroftKarp(g, r);
    // r[i] stores the left-side match for right-side node i
    for (int i = n; i < n + m; i++)
        if (r[i] != -1) cout << r[i] + 1 << " " << i - n + 1 << endl;
}
```

7 String

7.1 Trie

```
struct trienode {
    bool endmark;
    vector<int> child;
    trienode(int sz) {
        endmark = false;
        child.resize(sz, 0);
    }
};
struct trie {
    int sz;
    char fst;
    vector<trienode> nodes;
    trie(int alpha_sz, char alpha_start) {
        sz = alpha_sz;
        fst = alpha_start;
        // root at idx 0
        nodes.pb(trienode(sz));
    }
    void insert(string& s) {
        int cur = 0;
        for (char c : s) {
            if (!nodes[cur].child[c - fst]) {
                nodes[cur].child[c - fst] = nodes.size();
                nodes.pb(trienode(sz));
            }
            cur = nodes[cur].child[c - fst];
        }
        nodes[cur].endmark = true;
    }
    bool search(string& s) {
        int cur = 0;
        for (char c : s) {
```

```
            if (!nodes[cur].child[c - fst])
                return false;
            cur = nodes[cur].child[c - fst];
        }
        return nodes[cur].endmark;
    }
    bool erase(string& s) {
        int cur = 0;
        for (char c : s) {
            if (!nodes[cur].child[c - fst])
                return false;
            cur = nodes[cur].child[c - fst];
        }
        if (!nodes[cur].endmark) return false;
        nodes[cur].endmark = false;
        return true;
    }
};
```

7.2 Bit Trie

```
#define MAXN 100000011
int trie[MAXN][2], triesz[MAXN], nodes = 0, bitsz = 30;
void insert(int x) {
    int cur = 0;
    for (int i = bitsz; i >= 0; --i) {
        int child = x & (1 << i) ? 1 : 0;
        if (!trie[cur][child]) trie[cur][child] = ++nodes;
        triesz[cur]++;
        cur = trie[cur][child];
    }
    triesz[cur]++;
}
void erase(int x) {
    int cur = 0;
    for (int i = bitsz; i >= 0; --i) {
        int child = x & (1 << i) ? 1 : 0;
        if (!triesz[cur] || !trie[cur][child]) return;
        triesz[cur]--;
        cur = trie[cur][child];
    }
    triesz[cur]--;
}
// max xor with x
int get_max(int x) {
    int cur = 0, ans = 0;
    for (int i = bitsz; i >= 0; --i) {
        int child = x & (1 << i) ? 1 : 0;
        child = 1 - child; // remove for get_min()
        if (!trie[cur][child] || !triesz[trie[cur][child]]) child = 1 - child;
        cur = trie[cur][child];
        ans |= child << i;
    }
    return ans ^ x;
}
void deleteall(int root) {
    if (trie[root][0]) deleteall(trie[root][0]);
    if (trie[root][1]) deleteall(trie[root][1]);
    trie[root][0] = trie[root][1] = 0;
    triesz[root] = 0;
}
void cleanup() {
    nodes = 0;
    deleteall(0);
}
```

7.3 Z Algorithm

```
VI Z(string& s) {
    int n = s.size();
    VI z(n);
    int l = 0, r = 0;
    for (int i = 0; i < n; ++i) {
        if (i <= r) z[i] = min(z[i - l], r - i + 1);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) l = i, r = i + z[i]++;
    }
}
```

```
    return z;
}
```

7.4 KMP

```
vi pi(const string& s) {
    vi p(sz(s));
    rep(i, 1, sz(s)) {
        int g = p[i-1];
        while (g && s[i] != s[g]) g = p[g-1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}
vi match(const string& s, const string& pat) {
    vi p = pi(pat + '\0' + s), res;
    rep(i, sz(p)-sz(s), sz(p))
        if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
    return res;
}
```

7.5 Suffix Array

```
void suffix_array(string s){
    s+="$";
    int32_t n=s.size(),N=max(n,int32_t(260));
    vector<int32_t> sa(n),R(n);
    for(int32_t i=0;i<n;i++) sa[i]=i, R[i]=s[i];
    for(int32_t k=0;k<n;k?(k<=1):k++){
        vector<int32_t> nR(n),cnt(N);
        auto nsa=sa;
        for(int32_t i=0;i<n;i++) nsa[i]=(nsa[i]-k+n)%n, cnt[R[i]]++;
        for(int32_t i=1;i<N;i++) cnt[i]+=cnt[i-1];
        for(int32_t i=n-1;i>=0;i--) sa[--cnt[R[nsa[i]]]]=nsa[i];
        for(int32_t i=1,r=0;i<n;i++){
            r+=(R[sa[i]]!=R[sa[i-1]] || R[sa[i]+k]%n!=R[(sa[i-1]+k)%n]);
            nR[sa[i]]=r;
        }
        swap(R,nR);
    }
    vector<int32_t> lcp(n-1);
    for(int32_t i=0,k=0;i<n-1;i++){
        int32_t pi=R[i];
        int32_t j=sa[pi-1];
        while(i+k<n && j+k<n && s[i+k]==s[j+k]) k++;
        lcp[pi-1]=k;
        if(k) k--;
    }
}
```

7.6 Suffix Automaton

```
const int N = 3e5 + 9;
struct SuffixAutomaton {
    struct node {
        int len, link, firstpos;
        map<char, int> nxt;
    };
    int sz, last;
    vector<node> t;
    vector<int> terminal;
    vector<long long> dp;
    vector<vector<int>> g;
    SuffixAutomaton() {}
    SuffixAutomaton(int n) {
        t.resize(2 * n); terminal.resize(2 * n, 0);
        dp.resize(2 * n, -1); sz = 1;
        last = 0;
        g.resize(2 * n);
        t[0].len = 0; t[0].link = -1; t[0].firstpos = 0;
    }
    void extend(char c) {
        int p = last;
        if (t[p].nxt.count(c)) {
            int q = t[p].nxt[c];
```

```

    if (t[q].len == t[p].len +
1) {
        last = q;
        return;
    }
    int clone = sz++;
    t[clone] = t[q];
    t[clone].len = t[p].len + 1;
    t[q].link = clone;
    last = clone;
    while (p != -1 && t[p].nxt[c
] == q) {
        t[p].nxt[c] = clone;
        p = t[p].link;
    }
    return;
}
int cur = sz++;
t[cur].len = t[last].len + 1;
t[cur].firstpos = t[cur].len;
p = last;
while (p != -1 && !t[p].nxt.
count(c)) {
    t[p].nxt[c] = cur;
    p = t[p].link;
}
if (p == -1) t[cur].link = 0;
else {
    int q = t[p].nxt[c];
    if (t[p].len + 1 == t[q].len
) t[cur].link = q;
    else {
        int clone = sz++;
        t[clone] = t[q];
        t[clone].len = t[p].len
+ 1;
        while (p != -1 && t[p].
nxt[c] == q) {
            t[p].nxt[c] = clone;
            p = t[p].link;
        }
        t[q].link = t[cur].link
= clone;
    }
    last = cur;
}
void build_tree() {
    for (int i = 1; i < sz; i++) g[t
[i].link].push_back(i);
}
void build(string &s) {
    for (auto x: s) {
        extend(x);
        terminal[last] = 1;
    }
    build_tree();
}
long long cnt(int i) {
    if (dp[i] != -1) return dp[i];
    long long ret = terminal[i];
    for (auto &x: g[i]) ret += cnt(x
);
    return dp[i] = ret;
}
};

```

7.7 Manacher

```

array<vector<int>, 2> manacher(const
string& s) {
    int n = sz(s);
    array<vector<int>, 2> p = {vector<int>(
n+1), vector<int>(n)};
    rep(z, 0, 2) for (int i=0, l=0, r=0; i < n
; i++) {
        int t = r-i+!z;
        if (i<r) p[z][i] = min(t, p[z][l+t])
;
        int L = i-p[z][i], R = i+p[z][i]-!z;
        while (L>=1 && R+1<n && s[L-1] == s[
R+1])
            p[z][i]++, L--, R++;
        if (R>r) l=L, r=R;
    }
    return p;
}

```

7.8 Aho Corasick

```

struct AhoCorasick {
    enum {alpha = 26, first = 'A'}; //
change this!
    struct Node {
        // (nmatches is optional)
        int back, next[alpha], start = -1,
end = -1, nmatches = 0;
        Node(int v) { memset(next, v, sizeof
(next)); }
    };
    vector<Node> N;
    vi backp;
    void insert(string& s, int j) {
        assert(!s.empty());
        int n = 0;
        for (char c : s) {
            int& m = N[n].next[c - first];
            if (m == -1) { n = m = sz(N); N.
emplace_back(-1); }
            else n = m;
        }
        if (N[n].end == -1) N[n].start = j;
        backp.push_back(N[n].end);
        N[n].end = j;
        N[n].nmatches++;
    }
    AhoCorasick(vector<string>& pat) : N
(1, -1) {
        rep(i, 0, sz(pat)) insert(pat[i], i);
        N[0].back = sz(N);
        N.emplace_back(0);
        queue<int> q;
        for (q.push(0); !q.empty(); q.pop())
        {
            int n = q.front(), prev = N[n].
back;
            rep(i, 0, alpha) {
                int &ed = N[n].next[i], y = N[
prev].next[i];
                if (ed == -1) ed = y;
                else {
                    N[ed].back = y;
                    (N[ed].end == -1 ? N[ed].end :
backp[N[ed].start])
                    = N[y].end;
                    N[ed].nmatches += N[y].
nmatches;
                    q.push(ed);
                }
            }
        }
    }
    vi find(string word) {
        int n = 0;
        vi res; // ll count = 0;
        for (char c : word) {
            n = N[n].next[c - first];
            res.push_back(N[n].end);
            // count += N[n].nmatches;
        }
        return res;
    }
    vector<vi> findAll(vector<string>& pat
, string word) {
        vi r = find(word);
        vector<vi> res(sz(word));
        rep(i, 0, sz(word)) {
            int ind = r[i];
            while (ind != -1) {
                res[i - sz(pat[ind]) + 1].
push_back(ind);
                ind = backp[ind];
            }
        }
        return res;
    }
};

```

7.9 Min Lexic Rotation

```

int mnlex(string& s) {
    int n = s.size(), sz = 2 * n, l = 0, j
= 1, k = 0;
    while(l + k < sz && j + k < sz) {
        int a = ((l + k >= n) ? s[l + k - n]

```

```

        : s[l + k]);
        int b = ((j + k >= n) ? s[j + k - n]
        : s[j + k]);
        if(a == b) k++;
        else if(a > b) { // if maxlex then (
a<b)
            l = l + k + 1; k = 0;
            if(l <= j) { l = j + 1; }
        }
        else {
            j = j + k + 1;
            k = 0;
            if(j <= l) { j = l + 1; }
        }
    }
    return min(l, j);
}

```

7.10 Longest Palindrome

```

void lp(string& s) {
    string t = s;
    reverse(t.begin(), t.end());
    int n = s.size();
    Hash sh(s), th(t);
    auto f = [&](int L, int R) {
        auto ss = sh.geth(L, R);
        auto tt = th.geth(n - R + 1, n - L +
1);
        return (ss == tt);
    };
    vector<int> R(n, 1), L(n, 1);
    for (int i = n - 1, r = n - 1; i >= 0;
i--) {
        if (r < n - 1 && s[i] == s[r + 1]) {
            r++;
        }
        else {
            while (!f(i + 1, r + 1)) {
                r--;
            }
            R[i] = max(R[i], r - i + 1);
        }
    }
    for (int i = 0, l = 0; i < n; i++) {
        if (l > 0 && s[i] == s[l - 1]) {
            l--;
        }
        else {
            while (!f(l + 1, i + 1)) {
                l++;
            }
            L[i] = max(L[i], i - l + 1);
        }
    }
}

```

7.11 Wildcard Matching

```

//find occurrences of t in s where '?'s
are automatically matched with any
character
//res[i + m - 1] = sum_j=0 to m - 1 {s[i
+ j] * t[j] * (s[i + j] - t[j])^2
vector<int> string_matching(string &s,
string &t) {
    int n = s.size(), m = t.size();
    vector<int> s1(n), s2(n), s3(n);
    for(int i = 0; i < n; i++) s1[i] = s[i
] == '?' ? 0 : s[i] - 'a' + 1; //
assign any non zero number for non
'?'s
    for(int i = 0; i < n; i++) s2[i] = s1[
i] * s1[i];
    for(int i = 0; i < n; i++) s3[i] = s1[
i] * s2[i];
    vector<int> t1(m), t2(m), t3(m);
    for(int i = 0; i < m; i++) t1[i] = t[i
] == '?' ? 0 : t[i] - 'a' + 1;
    for(int i = 0; i < m; i++) t2[i] = t1[
i] * t1[i];
    for(int i = 0; i < m; i++) t3[i] = t1[
i] * t2[i];
    reverse(t1.begin(), t1.end());
    reverse(t2.begin(), t2.end());
    reverse(t3.begin(), t3.end());
    vector<int> s1t3 = multiply(s1, t3);
    vector<int> s2t2 = multiply(s2, t2);
    vector<int> s3t1 = multiply(s3, t1);

```

```
vector<int> res(n);
for(int i = 0; i < n; i++) res[i] =
    s1t3[i] - s2t2[i] * 2 + s3t1[i];
vector<int> oc;
for(int i = m - 1; i < n; i++) if(res[i]
    == 0) oc.push_back(i - m + 1);
return oc;
}
```

7.12 Hashing

```
const int SZ = int(1e6) + 9;
const int B1 = 151, B2 = 239;
const int mod1 = 127657753, mod2 =
    987654319;
vector<int> P1(SZ, 1), P2(SZ, 1);
void pow_cal() {
    for (int i = 1; i < SZ; i++) {
        P1[i] = (P1[i - 1] * B1) % mod1;
        P2[i] = (P2[i - 1] * B2) % mod2;
    }
}
class Hash {
public:
    int n;
    vector<int> H1, H2;
    Hash(const string& s) {
        n = s.size();
        H1.assign(n + 1, 0);
        H2.assign(n + 1, 0);
        for (int i = 1; i <= n; i++) {
            H1[i] = (H1[i - 1] + ((s[i - 1] -
                'a' + 1) * P1[i]) % mod1) % mod1;
            H2[i] = (H2[i - 1] + ((s[i - 1] -
                'a' + 1) * P2[i]) % mod2) % mod2;
        }
    }
    pair<int, int> geth(int l, int r) {
        int h1 = (((H1[r] - H1[l - 1] + mod1)
            ) % mod1) * P1[SZ - 1] % mod1;
        int h2 = (((H2[r] - H2[l - 1] + mod2)
            ) % mod2) * P2[SZ - 1] % mod2;
        return make_pair(h1, h2);
    }
    pair<int, int> geth() { return geth(1,
        n); }
};
```

7.13 2D Hash

```
struct Hashing {
    vector<vector<int>>> hs;
    vector<int> PWX, PWY;
    int n, m;
    static const int PX = 3731, PY = 2999,
        mod = 998244353;
    Hashing() {}
    Hashing(vector<string>& s) {
        n = (int)s.size(), m = (int)s[0].
            size();
        hs.assign(n + 1, vector<int>(m + 1,
            0));
        PWX.assign(n + 1, 1);
        PWY.assign(m + 1, 1);
        for (int i = 0; i < n; i++) PWX[i +
            1] = 1LL * PWX[i] * PX % mod;
        for (int i = 0; i < m; i++) PWY[i +
            1] = 1LL * PWY[i] * PY % mod;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                hs[i + 1][j + 1] = s[i][j] - 'a'
                    + 1;
            }
        }
        for (int i = 0; i <= n; i++) {
            for (int j = 0; j < m; j++) {
                hs[i][j + 1] = (hs[i][j + 1] + 1
                    LL * hs[i][j] * PY % mod) % mod;
            }
        }
        for (int i = 0; i < n; i++) {
            for (int j = 0; j <= m; j++) {
                hs[i + 1][j] = (hs[i + 1][j] + 1
                    LL * hs[i][j] * PX % mod) % mod;
            }
        }
    }
};
```

```
int get_hash(int x1, int y1, int x2,
    int y2) { // 1-indexed
    assert(1 <= x1 && x1 <= x2 && x2 <=
        n);
    assert(1 <= y1 && y1 <= y2 && y2 <=
        m);
    x1--;
    y1--;
    int dx = x2 - x1, dy = y2 - y1;
    return (1LL * (hs[x2][y2] - 1LL * hs
        [x2][y1] * PWY[dy] % mod + mod) %
        mod -
        1LL * (hs[x1][y2] - 1LL * hs
        [x1][y1] * PWY[dy] % mod + mod) %
        mod *
        PWX[dx] % mod +
        mod) %
        mod;
}
int get_hash() { return get_hash(1, 1,
    n, m); }
```

8 Numerical

8.1 FFT

```
using C = complex<double>;
using vd = vector<double>;
void fft(vector<C>& a) {
    int n = a.size(), L = 31 -
        __builtin_clz(n);
    static vector<complex<long double>>
        R(2, 1);
    static vector<C> rt(2, 1);
    static int kBuilt = 2;
    for (; kBuilt < n; kBuilt *= 2) {
        R.resize(2 * kBuilt);
        rt.resize(2 * kBuilt);
        auto x = polar(1.0L, acos(-1.0L))
            / kBuilt;
        for (int i = kBuilt; i < 2 *
            kBuilt; i++)
            rt[i] = R[i] = (i & 1) ? R[i
                / 2] * x : R[i / 2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; i++) rev[i] =
        (rev[i / 2] | (i & 1) << L) / 2;
    for (int i = 0; i < n; i++) if (i <
        rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 *
            k)
            for (int j = 0; j < k; j++)
            {
                // Hand-rolled complex
                multiply (~25% faster than rt[j+k]*a
                    [i+j+k])
                auto x = (double*)&rt[j
                    + k];
                auto y = (double*)&a[i +
                    j + k];
                C z(x[0] * y[0] - x[1] *
                    y[1],
                    x[0] * y[1] + x[1] *
                    y[0]);
                a[i + j + k] = a[i + j]
                    - z;
                a[i + j] += z;
            }
    }
    vector<ll> conv(const vector<ll>& a,
        const vector<ll>& b) {
        if (a.empty() || b.empty()) return
            {};
        vector<ll> res(a.size() + b.size() -
            1);
        int L = 32 - __builtin_clz((int)res.
            size()), n = 1 << L;
        vector<C> in(n), out(n);
        copy(a.begin(), a.end(), in.begin())
            ;
        for (int i = 0; i < (int)b.size(); i
            ++) in[i].imag(b[i]);
        fft(in);
        for (C& x : in) x *= x;
        for (int i = 0; i < n; i++) out[i] =
            in[i] & (n - 1) - conj(in[i]);
        fft(out);
        for (int i = 0; i < (int)res.size();
            i++) {
            res[i] = (ll)round(imag(out[i])
                / (4 * n));
        }
        return res;
    }
}
```

```
for (C& x : in) x *= x;
for (int i = 0; i < n; i++) out[i] =
    in[i] & (n - 1) - conj(in[i]);
fft(out);
for (int i = 0; i < (int)res.size();
    i++) {
    res[i] = (ll)round(imag(out[i])
        / (4 * n));
}
return res;
}
```

8.2 FFT With mod M

```
using C = complex<double>;
using vd = vector<double>;
void fft(vector<C>& a) {
    int n = a.size(), L = 31 -
        __builtin_clz(n);
    static vector<complex<long double>>
        R(2, 1);
    static vector<C> rt(2, 1);
    static int kBuilt = 2;
    for (; kBuilt < n; kBuilt *= 2) {
        R.resize(2 * kBuilt);
        rt.resize(2 * kBuilt);
        auto x = polar(1.0L, acos(-1.0L))
            / kBuilt;
        for (int i = kBuilt; i < 2 *
            kBuilt; i++)
            rt[i] = R[i] = (i & 1) ? R[i
                / 2] * x : R[i / 2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; i++) rev[i] =
        (rev[i / 2] | (i & 1) << L) / 2;
    for (int i = 0; i < n; i++) if (i <
        rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 *
            k)
            for (int j = 0; j < k; j++)
            {
                // Hand-rolled complex
                multiply (~25% faster than rt[j+k]*a
                    [i+j+k])
                auto x = (double*)&rt[j
                    + k];
                auto y = (double*)&a[i +
                    j + k];
                C z(x[0] * y[0] - x[1] *
                    y[1],
                    x[0] * y[1] + x[1] *
                    y[0]);
                a[i + j + k] = a[i + j]
                    - z;
                a[i + j] += z;
            }
    }
    vector<ll> conv(const vector<ll>& a,
        const vector<ll>& b) {
        if (a.empty() || b.empty()) return
            {};
        vector<ll> res(a.size() + b.size() -
            1);
        int L = 32 - __builtin_clz((int)res.
            size()), n = 1 << L;
        vector<C> in(n), out(n);
        copy(a.begin(), a.end(), in.begin())
            ;
        for (int i = 0; i < (int)b.size(); i
            ++) in[i].imag(b[i]);
        fft(in);
        for (C& x : in) x *= x;
        for (int i = 0; i < n; i++) out[i] =
            in[i] & (n - 1) - conj(in[i]);
        fft(out);
        for (int i = 0; i < (int)res.size();
            i++) {
            res[i] = (ll)round(imag(out[i])
                / (4 * n));
        }
        return res;
    }
}
```

8.3 NTT

```

const long long MOD = 998244353, root = 3;
void ntt(long long* x, long long* temp, long long* roots, int N, int skip) {
    if (N == 1) return;
    int n2 = N / 2;
    ntt(x, temp, roots, n2, skip * 2);
    ntt(x + skip, temp, roots, n2, skip * 2);
    for (int i = 0; i < N; i++) temp[i] = x[i * skip];
    for (int i = 0; i < n2; i++) {
        long long s = temp[2 * i], t = temp[2 * i + 1] * roots[skip * i];
        x[skip * i] = (s + t) % MOD;
        x[skip * (i + n2)] = ((s - t) % MOD + MOD) % MOD;
    }
}
void ntt(vector<long long>& x, bool inv = false) {
    long long e = modpow(root, (MOD - 1) / x.size(), MOD);
    if (inv) e = modpow(e, MOD - 2, MOD);
    vector<long long> roots(x.size(), 1), temp = roots;
    for (size_t i = 1; i < x.size(); i++) roots[i] = roots[i - 1] * e % MOD;
    ntt(&x[0], &temp[0], &roots[0], x.size(), 1);
}
vector<long long> conv(vector<long long> a, vector<long long> b) {
    int s = a.size() + b.size() - 1;
    if (s <= 0) return {};
    int L = s > 1 ? 32 - __builtin_clz(s - 1) : 0, n = 1 << L;
    if (s <= 200) {
        vector<long long> c(s);
        for (size_t i = 0; i < a.size(); i++) {
            for (size_t j = 0; j < b.size(); j++) {
                c[i + j] = (c[i + j] + a[i] * b[j]) % MOD;
            }
        }
        return c;
    }
    a.resize(n); ntt(a);
    b.resize(n); ntt(b);
    vector<long long> c(n);
    long long d = modpow(n, MOD - 2, MOD);
    for (int i = 0; i < n; i++) c[i] = a[i] * b[i] % MOD * d % MOD;
    ntt(c, true);
    c.resize(s);
    return c;
}
vector<long long> mul(const vector<long long>& a, const vector<long long>& b, int n) {
    vector<long long> c = conv(a, b);
    if ((int)c.size() > n + 1) c.resize(n + 1);
    return c;
}
vector<long long> power(vector<long long> > f, long long m, int n) {
    vector<long long> r = {1};
    if ((int)f.size() > n + 1) f.resize(n + 1);
    for (; m > 0; m >= 1, f = mul(f, f, n)) {
        if (m & 1) r = mul(r, f, n);
        return r;
    }
}

```

8.4 FWHT

```

void FST(vector<int>& a, bool inv) {
    for (int n = a.size(), step = 1; step < n; step *= 2) {
        for (int i = 0; i < n; i += 2 * step)

```

```

        for(int j = i; j < i+step; j++) {
            int &u = a[j], &v = a[j + step]; tie(u, v) = inv ? pair<int,int>(v - u, u) : pair<int,int>(v, u + v); // AND
            // inv ? pair<int,int>(v, u - v) : pair<int,int>(u + v, u); // OR
            // include-line // pair<int,int>(u + v, u - v); // XOR
            // include-line
        }
        // if (inv) for (int& x : a) x /= a.size(); // XOR only
    }
}
vector<int> conv(vector<int> a, vector<int> b) {
    int n = a.size();
    FST(a, 0); FST(b, 0);
    for(int i = 0; i < n; i++) a[i] *= b[i];
    FST(a, 1); return a;
}

```

9 Linear Algebra

9.1 The Matrix Class

```

struct Mat {
    int n, m;
    vector<vector<int>> a;
    Mat() {}
    Mat(int _n, int _m) {n = _n; m = _m; a.assign(n, vector<int>(m, 0));}
    Mat(vector<vector<int>> > v) {n = v.size(); m = n ? v[0].size() : 0; a = v;}
    inline void make_unit() {assert(n == m); for (int i = 0; i < n; i++) {for (int j = 0; j < n; j++) a[i][j] = i == j;}}
    inline Mat operator + (const Mat &b) {assert(n == b.n && m == b.m); Mat ans = Mat(n, m); for(int i = 0; i < n; i++) {for(int j = 0; j < m; j++) {ans.a[i][j] = (a[i][j] + b.a[i][j]) % mod;}} return ans;}
    inline Mat operator - (const Mat &b) {assert(n == b.n && m == b.m); Mat ans = Mat(n, m); for(int i = 0; i < n; i++) {for(int j = 0; j < m; j++) {ans.a[i][j] = (a[i][j] - b.a[i][j] + mod) % mod;}} return ans;}
    inline Mat operator * (const Mat &b) {assert(m == b.n); Mat ans = Mat(n, b.m); for(int i = 0; i < n; i++) {for(int j = 0; j < b.m; j++) {for(int k = 0; k < m; k++) {ans.a[i][j] = (ans.a[i][j] + 1LL * a[i][k] * b.a[k][j] % mod) % mod;}} return ans;}
    inline Mat pow(long long k) {assert(n == m); Mat ans(n, n), t = a; ans.make_unit();

```

```

    while (k) {
        if (k & 1) ans = ans * t;
        t = t * t;
        k >>= 1;
    }
    return ans;
}
inline Mat& operator += (const Mat& b) {return *this = (*this) + b;}
inline Mat& operator -= (const Mat& b) {return *this = (*this) - b;}
inline Mat& operator *= (const Mat& b) {return *this = (*this) * b;}
inline bool operator == (const Mat& b) {return a == b.a;}
inline bool operator != (const Mat& b) {return a != b.a;}
};

```

9.2 Gaussian Elimination

```

int Gauss(Mat a, vector<int> &ans) {
    int n = a.n, m = a.m - 1;
    vector<int> pos(m, -1);
    int free_var = 0;
    const long long MODSQ = (long long)mod * mod;
    int rank = 0;
    for (int col = 0, row = 0; col < m && row < n; col++) {
        int mx = row;
        for (int k = row; k < n; k++) if (a.a[k][col] > a.a[mx][col]) mx = k;
        if (a.a[mx][col] == 0) continue;
        for (int j = col; j <= m; j++) swap(a.a[mx][j], a.a[row][j]);
        pos[col] = row;
        int inv = power(a.a[row][col], mod - 2);
        for (int i = 0; i < n && inv; i++) {
            if (i != row && a.a[i][col]) {
                int x = (1LL * a.a[i][col] * inv) % mod;
                for (int j = col; j <= m && x; j++) {
                    if (a.a[row][j]) a.a[i][j] = (MODSQ + a.a[i][j] - 1LL * a.a[row][j] * x) % mod;
                }
            }
        }
        row++; ++rank;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++) {
        if (pos[i] == -1) free_var++;
        else ans[i] = (1LL * a.a[pos[i]][m] * power(a.a[pos[i]][i], mod - 2)) % mod;
    }
    for (int i = 0; i < n; i++) {
        long long val = 0;
        for (int j = 0; j < m; j++) val = (val + 1LL * ans[j] * a.a[i][j]) % mod;
        if (val != a.a[i][m]) return -1;
    }
    return free_var;
}

```

9.3 Determinant

```

int Det(Mat a) {
    assert(a.n == a.m);
    int n = a.n;
    const long long MODSQ = (long long)mod * mod;
    int det = 1;
    for (int col = 0, row = 0; col < n && row < n; col++) {
        int mx = row;
        for (int k = row; k < n; k++) if (a.a[k][col] > a.a[mx][col]) mx = k;
        if (a.a[mx][col] == 0) {det = 0; continue;}
        for (int j = col; j < n; j++) swap(a.a[mx][j], a.a[row][j]);
    }
}

```

```

if (row != mx) det = (det == 0 ? 0 :
    mod - det);
det = 1LL * det * a.a[row][col] %
mod;
int inv = power(a.a[row][col], mod -
2);
for (int i = 0; i < n && inv; i++) {
    if (i != row && a.a[i][col]) {
        int x = (1LL * a.a[i][col] * inv
        ) % mod;
        for (int j = col; j < n && x; j
        ++){
            if (a.a[row][j]) a.a[i][j] = (
MODSQ + a.a[i][j] - 1LL * a.a[row][
j] * x) % mod;
        }
    }
    row++;
}
return det;
}

```

9.4 Matrix Inverse

```

bool Inv(Mat a, Mat &ans) {
    assert(a.n == a.m);
    int n = a.n;
    vector<int> col(n);
    Mat tmp(n, n); tmp.make_unit();
    for (int i = 0; i < n; i++) col[i] = i
    ;
    for (int i = 0; i < n; i++) {
        int r = i, c = i;
        for (int j = i; j < n; j++)
            for (int k = i; k < n; k++)
                if (a.a[j][k]) { r = j; c = k;
                goto found; }
        return false;
    found:
        a.a[i].swap(a.a[r]); tmp.a[i].swap(
        tmp.a[r]);
        for (int j = 0; j < n; j++) { swap(a
        .a[j][i], a.a[j][c]); swap(tmp.a[j
        ][i], tmp.a[j][c]); }
        swap(col[i], col[c]);
        int v = power(a.a[i][i], mod - 2);
        for (int j = i + 1; j < n; j++) {
            int f = 1LL * a.a[j][i] * v % mod;
            a.a[j][i] = 0;
            for (int k = i + 1; k < n; k++) {
                a.a[j][k] -= 1LL * f * a.a[i][k]
                % mod;
                if (a.a[j][k] < 0) a.a[j][k] +=
                mod;
            }
            for (int k = 0; k < n; k++) {
                tmp.a[j][k] -= 1LL * f * tmp.a[i
                ][k] % mod;
                if (tmp.a[j][k] < 0) tmp.a[j][k]
                += mod;
            }
        }
        for (int j = i + 1; j < n; j++) a.a[
        i][j] = 1LL * a.a[i][j] * v % mod;
        for (int j = 0; j < n; j++) tmp.a[i
        ][j] = 1LL * tmp.a[i][j] * v % mod;
        a.a[i][i] = 1;
    }
}

for (int i = n - 1; i > 0; --i) {
    for (int j = 0; j < i; j++) {
        int v = a.a[j][i];
        for (int k = 0; k < n; k++) {
            tmp.a[j][k] -= 1LL * v * tmp.a[i
            ][k] % mod;
            if (tmp.a[j][k] < 0) tmp.a[j][k]
            += mod;
        }
    }
}
ans = Mat(n, n);
for (int i = 0; i < n; i++)
    for (int j = 0; j < n; j++)
        ans.a[col[i]][col[j]] = tmp.a[i][j
        ];
return true;
}

```

9.5 Xor Basis

```

const int LOG_K = 60; // Adjust as
needed
struct XorBasis {
    vector<int64_t> basis;
    int64_t N = 0, tmp = 0;
    void add(int64_t x) {
        N++;
        tmp |= x;
        for (auto& i : basis) {
            x = min(x, x ^ i);
        }
        if (!x) return;
        for (auto& i : basis) {
            if ((i ^ x) < i) {
                i ^= x;
            }
        }
        basis.push_back(x);
        sort(basis.begin(), basis.end());
    }
    int64_t size() { return (int64_t)basis
    .size(); }
    void clear() {
        N = 0;
        tmp = 0;
        basis.clear();
    }
    bool possible(int64_t x) {
        for (auto& i : basis) {
            x = min(x, x ^ i);
        }
        return !x;
    }
    int64_t maxxor(int64_t x = 0) {
        for (auto& i : basis) {
            x = max(x, x ^ i);
        }
        return x;
    }
    int64_t minxor(int64_t x = 0) {
        for (auto& i : basis) {
            x = min(x, x ^ i);
        }
        return x;
    }
    int64_t cntxor(int64_t x) {
        if (!possible(x)) return 0;
        return (1LL << (N - size()));
    }
    int64_t sumOfAll() {
        int64_t ans = tmp * (1LL << (N - 1))
        ;
        return ans;
    }
    int64_t kth(int64_t k) {
        int64_t sz = size();
        if (k > (1LL << sz)) return -1;
        k--;
        int64_t ans = 0;
        for (int64_t i = 0; i < sz; i++) {
            if (k >= i & 1) {
                ans ^= basis[i];
            }
        }
        return ans;
    }
};

```

10 Geometry

10.1 Full Template

```

const int N = 3e5 + 9;
const double inf = 1e100;
const double PI = acos((double)-1.0);
const double eps = 1e-9;
// Returns -1 for negative, 0 for zero,
1 for positive (with epsilon
tolerance)
int sign(double x) { return (x > eps) -
(x < -eps); }
struct PT {
    double x, y;
    PT() { x = 0, y = 0; }

```

```

PT(double x, double y) : x(x), y(y) {}
PT(const PT &p) : x(p.x), y(p.y) {}
PT operator + (const PT &a) const {
    return PT(x + a.x, y + a.y); }
PT operator - (const PT &a) const {
    return PT(x - a.x, y - a.y); }
PT operator * (const double a) const {
    return PT(x * a, y * a); }
friend PT operator * (const double &a,
const PT &b) { return PT(a * b.x,
a * b.y); }
PT operator / (const double a) const {
    return PT(x / a, y / a); }
bool operator == (PT a) const { return
sign(a.x - x) == 0 && sign(a.y - y
) == 0; }
bool operator != (PT a) const { return
!(*this == a); }
bool operator < (PT a) const { return
sign(a.x - x) == 0 ? y < a.y : x <
a.x; }
bool operator > (PT a) const { return
sign(a.x - x) == 0 ? y > a.y : x >
a.x; }
double norm() { return sqrt(x * x + y
* y); }
double norm2() { return x * x + y * y;
}
PT perp() { return PT(-y, x); }
double arg() { return atan2(y, x); }
// Returns a vector with norm r and
having same direction
PT truncate(double r) {
    double k = norm();
    if (!sign(k)) return *this;
    r /= k;
    return PT(x * r, y * r);
}
};
istream &operator >> (istream &in, PT &p
) { return in >> p.x >> p.y; }
ostream &operator << (ostream &out, PT &
p) { return out << "(" << p.x << ",
" << p.y << ")"; }
inline double dot(PT a, PT b) { return a
.x * b.x + a.y * b.y; }
inline double dist2(PT a, PT b) { return
dot(a - b, a - b); }
inline double dist(PT a, PT b) { return
sqrt(dot(a - b, a - b)); }
inline double cross(PT a, PT b) { return
a.x * b.y - a.y * b.x; }
inline double cross2(PT a, PT b, PT c) {
    return cross(b - a, c - a); }
inline int orientation(PT a, PT b, PT c)
{ return sign(cross(b - a, c - a))
; }
PT perp(PT a) { return PT(-a.y, a.x); }
PT rotateccw90(PT a) { return PT(-a.y, a
.x); }
PT rotatecw90(PT a) { return PT(a.y, -a.
x); }
PT rotateccw(PT a, double t) { return PT
(a.x * cos(t) - a.y * sin(t), a.x *
sin(t) + a.y * cos(t)); }
PT rotatecw(PT a, double t) { return PT(
a.x * cos(t) + a.y * sin(t), -a.x *
sin(t) + a.y * cos(t)); }
double SQ(double x) { return x * x; }
double rad_to_deg(double r) { return (r
* 180.0 / PI); }
double deg_to_rad(double d) { return (d
* PI / 180.0); }
// Returns angle between vectors a and b
in radians
double get_angle(PT a, PT b) {
    double costheta = dot(a, b) / a.norm()
/ b.norm();
    return acos(max((double)-1.0, min((
double)1.0, costheta)));
}
struct line {
    PT a, b; // goes through points a and
b
    PT v; double c; // line form:
direction vec [cross] (x, y) = c
    line() {}
    // Direction vector v and offset c

```

```

line(PT v, double c) : v(v), c(c) {
    auto p = get_points();
    a = p.first; b = p.second;
}
// Equation ax + by + c = 0
line(double _a, double _b, double _c)
: v({b, -a}), c(-c) {
    auto p = get_points();
    a = p.first; b = p.second;
}
// Goes through points p and q
line(PT p, PT q) : v(q - p), c(cross(v
, p)), a(p), b(q) {}
// Extract any two points from this
line
pair<PT, PT> get_points() {
    PT p, q;
    double a_coef = -v.y, b_coef = v.x;
    // ax + by = c
    if (sign(a_coef) == 0) {
        p = PT(0, c / b_coef);
        q = PT(1, c / b_coef);
    } else if (sign(b_coef) == 0) {
        p = PT(c / a_coef, 0);
        q = PT(c / a_coef, 1);
    } else {
        p = PT(0, c / b_coef);
        q = PT(1, (c - a_coef) / b_coef);
    }
    return {p, q};
}
// Returns coefficients for ax + by +
c = 0
array<double, 3> get_abc() {
    double a_coef = -v.y, b_coef = v.x;
    return {a_coef, b_coef, -c};
}
// 1 if on the left, -1 if on the
right, 0 if on the line
int side(PT p) { return sign(cross(v,
p) - c); }
// Line that is perpendicular to this
and goes through point p
line perpendicular_through(PT p) {
    return {p, p + perp(v)}; }
// Translate the line by vector t
line translate(PT t) { return {v, c +
cross(v, t)}; }
// Compare two points by orthogonal
projection on this line
bool cmp_by_projection(PT p, PT q) {
    return dot(v, p) < dot(v, q); }
line shift_left(double d) {
    PT z = v.perp().truncate(d);
    return line(a + z, b + z);
}
};
// Minimum distance from point c to line
through a and b
double dist_from_point_to_line(PT a, PT
b, PT c) {
    return fabs(cross(b - a, c - a) / (b -
a).norm());
}
// Returns true if point p is on line
segment ab
bool is_point_on_seg(PT a, PT b, PT p) {
    if (fabs(cross(p - b, a - b)) < eps) {
        if (p.x < min(a.x, b.x) - eps || p.x
> max(a.x, b.x) + eps) return
false;
        if (p.y < min(a.y, b.y) - eps || p.y
> max(a.y, b.y) + eps) return
false;
        return true;
    }
    return false;
}
// Minimum distance point from point c
to segment ab that lies on segment
ab
PT project_from_point_to_seg(PT a, PT b,
PT c) {
    double r = dist2(a, b);
    if (sign(r) == 0) return a;
    r = dot(c - a, b - a) / r;
    if (r < 0) return a;
    if (r > 1) return b;
    return a + (b - a) * r;
}
// Minimum distance from point c to
segment ab
double dist_from_point_to_seg(PT a, PT b
, PT c) {
    return dist(c,
project_from_point_to_seg(a, b, c))
;
}
// 0 if not parallel, 1 if parallel, 2
if collinear
int is_parallel(PT a, PT b, PT c, PT d)
{
    double k = fabs(cross(b - a, d - c));
    if (k < eps) {
        if (fabs(cross(a - b, a - c)) < eps
&& fabs(cross(c - d, c - a)) < eps)
return 2;
        else return 1;
    } else return 0;
}
// 1 if point is ccw to the line, 2 if
point is cw to the line, 3 if point
is on the line
int point_line_relation(PT a, PT b, PT p
) {
    int c = sign(cross(p - a, b - a));
    if (c < 0) return 1;
    if (c > 0) return 2;
    return 3;
}
// Intersection point between line ab
and line cd assuming unique
intersection exists
bool line_line_intersection(PT a, PT b,
PT c, PT d, PT &ans) {
    double a1 = a.y - b.y, b1 = b.x - a.x,
c1 = cross(a, b);
    double a2 = c.y - d.y, b2 = d.x - c.x,
c2 = cross(c, d);
    double det = a1 * b2 - a2 * b1;
    if (det == 0) return 0;
    ans = PT((b1 * c2 - b2 * c1) / det, (
c1 * a2 - a1 * c2) / det);
    return 1;
}
// Intersection point between segment ab
and segment cd assuming unique
intersection exists
bool seg_seg_intersection(PT a, PT b, PT
c, PT d, PT &ans) {
    double oa = cross2(c, d, a), ob =
cross2(c, d, b);
    double oc = cross2(a, b, c), od =
cross2(a, b, d);
    if (oa * ob < 0 && oc * od < 0) {
        ans = (a * ob - b * oa) / (ob - oa);
        return 1;
    } else return 0;
}
// Custom change: Boolean intersection
check (handles collinear overlap)
bool seg_seg_intersection(PT a, PT b, PT
c, PT d) {
    double oa = cross2(c, d, a), ob =
cross2(c, d, b);
    double oc = cross2(a, b, c), od =
cross2(a, b, d);
    if (oa * ob < 0 && oc * od < 0) return
true;
    // Checking orientation for very
precise calculation
    if (oa == 0 && is_point_on_seg(c, d, a
)) return true;
    if (ob == 0 && is_point_on_seg(c, d, b
)) return true;
    if (oc == 0 && is_point_on_seg(a, b, c
)) return true;
    if (od == 0 && is_point_on_seg(a, b, d
)) return true;
    return false;
}
// 0 if do not intersect, 1 if proper
intersect, 2 if segment intersect
int seg_line_relation(PT a, PT b, PT c,
PT d) {
    double p = cross2(c, d, a);
    double q = cross2(c, d, b);
    if (sign(p) == 0 && sign(q) == 0)
return 2;
    else if (p * q < 0) return 1;
    else return 0;
}
// Intersection between segment ab and
line cd assuming unique intersection
exists
bool seg_line_intersection(PT a, PT b,
PT c, PT d, PT &ans) {
    bool k = seg_line_relation(a, b, c, d)
;
    assert(k != 2);
    if (k) line_line_intersection(a, b, c,
d, ans);
    return k;
}
// Minimum distance from segment ab to
segment cd
double dist_from_seg_to_seg(PT a, PT b,
PT c, PT d) {
    PT dummy;
    if (seg_seg_intersection(a, b, c, d,
dummy)) return 0.0;
    else return min({
        dist_from_point_to_seg(a, b, c),
        dist_from_point_to_seg(a, b, d),

        dist_from_point_to_seg(c, d, a),
        dist_from_point_to_seg(c, d, b)});
}
struct circle {
    PT p; double r;
    circle() {}
    circle(PT _p, double _r) : p(_p), r(_r)
{}
    circle(double x, double y, double _r) :
p(PT(x, y)), r(_r) {} // center (x
, y) and radius r
    // Circumcircle of a triangle (three
points must be unique)
    circle(PT a, PT b, PT c) {
        b = (a + b) * 0.5;
        c = (a + c) * 0.5;
        line_line_intersection(b, b +
rotatecw90(a - b), c, c +
rotatecw90(a - c), p);
        r = dist(a, p);
    }
    // Inscribed circle of a triangle (
pass a bool just to differentiate
from circumcircle)
    circle(PT a, PT b, PT c, bool t) {
        line u, v;
        double m = atan2(b.y - a.y, b.x - a.
x), n = atan2(c.y - a.y, c.x - a.x)
;
        u.a = a;
        u.b = u.a + (PT(cos((n + m)/2.0),
sin((n + m)/2.0)));
        v.a = b;
        m = atan2(a.y - b.y, a.x - b.x), n =
atan2(c.y - b.y, c.x - b.x);
        v.b = v.a + (PT(cos((n + m)/2.0),
sin((n + m)/2.0)));
        line_line_intersection(u.a, u.b, v.a
, v.b, p);
        r = dist_from_point_to_seg(a, b, p);
    }
    bool operator == (circle v) { return p
== v.p && sign(r - v.r) == 0; }
    double area() { return PI * r * r; }
    double circumference() { return 2.0 *
PI * r; }
};
// 0 if outside, 1 if on circumference,
2 if inside circle
int circle_point_relation(PT p, double r
, PT b) {
    double d = dist(p, b);
    if (sign(d - r) < 0) return 2;
    if (sign(d - r) == 0) return 1;
    return 0;
}
// 0 if outside, 1 if on circumference,
2 if inside circle
int circle_line_relation(PT p, double r,

```

```

    PT a, PT b) {
    double d = dist_from_point_to_line(a,
    b, p);
    if (sign(d - r) < 0) return 2;
    if (sign(d - r) == 0) return 1;
    return 0;
}
// Compute intersection of line through
// a and b with circle centered at c
// with radius r > 0
vector<PT> circle_line_intersection(PT c
, double r, PT a, PT b) {
    vector<PT> ret;
    b = b - a; a = a - c;
    double A = dot(b, b), B = dot(a, b);
    double C = dot(a, a) - r * r, D = B *
    B - A * C;
    if (D < -eps) return ret;
    ret.push_back(c + a + b * (-B + sqrt(D
    + eps)) / A);
    if (D > eps) ret.push_back(c + a + b *
    (-B - sqrt(D)) / A);
    return ret;
}
// 5 - outside and do not intersect, 4 -
// intersect outside in one point
// 3 - intersect in 2 points, 2 -
// intersect inside in one point, 1 -
// inside and do not intersect
int circle_circle_relation(PT a, double
r, PT b, double R) {
    double d = dist(a, b);
    if (sign(d - r - R) > 0) return 5;
    if (sign(d - r - R) == 0) return 4;
    double l = fabs(r - R);
    if (sign(d - r - R) < 0 && sign(d - l)
    > 0) return 3;
    if (sign(d - l) == 0) return 2;
    if (sign(d - l) < 0) return 1;
    assert(0); return -1;
}
vector<PT> circle_circle_intersection(PT
a, double r, PT b, double R) {
    if (a == b && sign(r - R) == 0) return
    {PT(1e18, 1e18)};
    vector<PT> ret;
    double d = sqrt(dist2(a, b));
    if (d > r + R || d + min(r, R) < max(r
    , R)) return ret;
    double x = (d * d - R * R + r * r) /
    (2 * d);
    double y = sqrt(r * r - x * x);
    PT v = (b - a) / d;
    ret.push_back(a + v * x + rotateccw90(
    v) * y);
    if (y > 0) ret.push_back(a + v * x -
    rotateccw90(v) * y);
    return ret;
}
// Returns two circle c1, c2 through
// points a, b and of radius r
// 0 if there is no such circle, 1 if
// one circle, 2 if two circles
int get_circle(PT a, PT b, double r,
circle &c1, circle &c2) {
    vector<PT> v =
    circle_circle_intersection(a, r, b,
    r);
    int t = v.size();
    if (!t) return 0;
    c1.p = v[0], c1.r = r;
    if (t == 2) c2.p = v[1], c2.r = r;
    return t;
}
// Returns area of intersection between
// two circles
double circle_circle_area(PT a, double
r1, PT b, double r2) {
    double d = (a - b).norm();
    if (r1 + r2 < d + eps) return 0;
    if (r1 + d < r2 + eps) return PI * r1
    * r1;
    if (r2 + d < r1 + eps) return PI * r2
    * r2;
    double theta_1 = acos((r1 * r1 + d * d
    - r2 * r2) / (2 * r1 * d));
    double theta_2 = acos((r2 * r2 + d * d
    - r1 * r1) / (2 * r2 * d));

```

```

    return r1 * r1 * (theta_1 - sin(2 *
    theta_1) / 2.0) + r2 * r2 * (
    theta_2 - sin(2 * theta_2) / 2.0);
}
double area_of_triangle(PT a, PT b, PT c
) {
    return fabs(cross(b - a, c - a) * 0.5)
    ;
}
// -1 if strictly inside, 0 if on the
// polygon, 1 if strictly outside
int is_point_in_triangle(PT a, PT b, PT
c, PT p) {
    if (sign(cross(b - a, c - a)) < 0)
    swap(b, c);
    int c1 = sign(cross(b - a, p - a));
    int c2 = sign(cross(c - b, p - b));
    int c3 = sign(cross(a - c, p - c));
    if (c1 < 0 || c2 < 0 || c3 < 0) return
    -1;
    if (c1 + c2 + c3 != 3) return 0;
    return 1;
}
double area(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans +=
    cross(p[i], p[(i + 1) % n]);
    return fabs(ans) * 0.5;
}
// 0 if cw, 1 if ccw
bool get_direction(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans +=
    cross(p[i], p[(i + 1) % n]);
    if (sign(ans) > 0) return 1;
    return 0;
}
// Returns the convex hull of the set of
// points p (Monotone Chain)
vector<PT> convex_hull(vector<PT> &p) {
    if (p.size() <= 1) return p;
    vector<PT> v = p;
    sort(v.begin(), v.end());
    vector<PT> up, dn;
    for (auto& pt : v) {
        while (up.size() > 1 && orientation(
        up[up.size() - 2], up.back(), pt)
        >= 0) up.pop_back();
        while (dn.size() > 1 && orientation(
        dn[dn.size() - 2], dn.back(), pt)
        <= 0) dn.pop_back();
        up.push_back(pt);
        dn.push_back(pt);
    }
    v = dn;
    if (v.size() > 1) v.pop_back();
    reverse(up.begin(), up.end());
    up.pop_back();
    for (auto& pt : up) v.push_back(pt);
    if (v.size() == 2 && v[0] == v[1]) v.
    pop_back();
    return v;
}
// Checks if convex or not
bool is_convex(vector<PT> &p) {
    bool s[3] = {0, 0, 0};
    int n = p.size();
    for (int i = 0; i < n; i++) {
        int j = (i + 1) % n;
        int k = (j + 1) % n;
        s[sign(cross(p[j] - p[i], p[k] - p[i]
        )) + 1] = 1;
        if (s[0] && s[2]) return 0;
    }
    return 1;
}
// -1 if strictly inside, 0 if on the
// polygon, 1 if strictly outside (O(
// log n))
int is_point_in_convex(vector<PT> &p,
const PT& x) {
    int n = p.size(); assert(n >= 3);
    int a = orientation(p[0], p[1], x), b
    = orientation(p[0], p[n - 1], x);
    if (a < 0 || b > 0) return 1;
    int l = 1, r = n - 1;
    while (l + 1 < r) {
        int mid = (l + r) >> 1;

```

```

        if (orientation(p[0], p[mid], x) >=
        0) l = mid;
        else r = mid;
    }
    int k = orientation(p[1], p[r], x);
    if (k <= 0) return -k;
    if (l == 1 && a == 0) return 0;
    if (r == n - 1 && b == 0) return 0;
    return -1;
}
bool is_point_on_polygon(vector<PT> &p,
const PT& z) {
    int n = p.size();
    for (int i = 0; i < n; i++) {
        if (is_point_on_seg(p[i], p[(i + 1)
        % n], z)) return 1;
    }
    return 0;
}
// Returns 1e9 if the point is on the
// polygon (O(n))
int winding_number(vector<PT> &p, const
PT& z) {
    if (is_point_on_polygon(p, z)) return
    1e9;
    int n = p.size(), ans = 0;
    for (int i = 0; i < n; i++) {
        int j = (i + 1) % n;
        bool below = p[i].y < z.y;
        if (below != (p[j].y < z.y)) {
            auto orient = orientation(z, p[j],
            p[i]);
            if (orient == 0) return 0;
            if (below == (orient > 0)) ans +=
            below ? 1 : -1;
        }
    }
    return ans;
}
// -1 if strictly inside, 0 if on the
// polygon, 1 if strictly outside
int is_point_in_polygon(vector<PT> &p,
const PT& z) {
    int k = winding_number(p, z);
    return k == 1e9 ? 0 : k == 0 ? 1 : -1;
}
// id of the vertex having maximum dot
// product with z (O(log n))
int extreme_vertex(vector<PT> &p, const
PT& z, const int top) {
    int n = p.size();
    if (n == 1) return 0;
    double ans = dot(p[0], z); int id = 0;
    if (dot(p[top], z) > ans) ans = dot(p[
    top], z), id = top;
    int l = 1, r = top - 1;
    while (l < r) {
        int mid = (l + r) >> 1;
        if (dot(p[mid + 1], z) >= dot(p[mid
        ], z)) l = mid + 1;
        else r = mid;
    }
    if (dot(p[1], z) > ans) ans = dot(p[1
    ], z), id = 1;
    l = top + 1, r = n - 1;
    while (l < r) {
        int mid = (l + r) >> 1;
        if (dot(p[(mid + 1) % n], z) >= dot(
        p[mid], z)) l = mid + 1;
        else r = mid;
    }
    l %= n;
    if (dot(p[1], z) > ans) ans = dot(p[1
    ], z), id = 1;
    return id;
}
pair<PT, int> point_poly_tangent(vector<
PT> &p, PT Q, int dir, int l, int r
) {
    while (r - l > 1) {
        int mid = (l + r) >> 1;
        bool pvs = orientation(Q, p[mid], p[
        mid - 1]) != -dir;
        bool nxt = orientation(Q, p[mid], p[
        mid + 1]) != -dir;
        if (pvs && nxt) return {p[mid], mid
        };
        if (!(pvs || nxt)) {

```

```

    auto p1 = point_poly_tangent(p, Q,
    dir, mid + 1, r);
    auto p2 = point_poly_tangent(p, Q,
    dir, l, mid - 1);
    return orientation(Q, p1.first, p2
    .first) == dir ? p1 : p2;
}
if (!pvs) {
    if (orientation(Q, p[mid], p[l])
    == dir) r = mid - 1;
    else if (orientation(Q, p[l], p[r]
    ]) == dir) r = mid - 1;
    else l = mid + 1;
}
if (!nxt) {
    if (orientation(Q, p[mid], p[l])
    == dir) l = mid + 1;
    else if (orientation(Q, p[l], p[r]
    ]) == dir) r = mid - 1;
    else l = mid + 1;
}
}
pair<PT, int> ret = {p[l], l};
for (int i = l + 1; i <= r; i++) {
    ret = orientation(Q, ret.first, p[i]
    ]) != dir ? make_pair(p[i], i) :
    ret;
}
return ret;
}
// (ccw, cw) tangents from a point that
// is outside this convex polygon
pair<int, int>
tangents_from_point_to_polygon(
    vector<PT> &p, PT Q) {
    int ccw = point_poly_tangent(p, Q, 1,
    0, (int)p.size() - 1).second;
    int cw = point_poly_tangent(p, Q, -1,
    0, (int)p.size() - 1).second;
    return make_pair(ccw, cw);
}
// Minimum distance from a point to a
// convex polygon (assumes strictly
// outside)
double dist_from_point_to_polygon(vector
<PT> &p, PT z) {
    double ans = inf;
    int n = p.size();
    if (n <= 3) {
        for (int i = 0; i < n; i++) ans = min
        (ans, dist_from_point_to_seg(p[i],
        p[(i + 1) % n], z));
        return ans;
    }
    auto [r, l] =
        tangents_from_point_to_polygon(p, z
        );
    if (l > r) r += n;
    while (l < r) {
        int mid = (l + r) >> 1;
        double left = dist2(p[mid % n], z),
        right = dist2(p[(mid + 1) % n], z);
        ans = min({ans, left, right});
        if (left < right) r = mid;
        else l = mid + 1;
    }
    ans = sqrt(ans);
    ans = min(ans, dist_from_point_to_seg(
    p[l % n], p[(l + 1) % n], z));
    ans = min(ans, dist_from_point_to_seg(
    p[l % n], p[(l - 1 + n) % n], z));
    return ans;
}
// Minimum distance from convex polygon
// p to line ab (O(log n))
double dist_from_polygon_to_line(vector<
PT> &p, PT a, PT b, int top) {
    PT orth = (b - a).perp();
    if (orientation(a, b, p[0]) > 0) orth
    = (a - b).perp();
    int id = extreme_vertex(p, orth, top);
    if (dot(p[id] - a, orth) > 0) return
    0.0; // Intersects
    return dist_from_point_to_line(a, b, p
    [id]); // Does not intersect
}
// Minimum distance from a convex
// polygon to another convex polygon (O

```

```

(n log n))
double dist_from_polygon_to_polygon(
    vector<PT> &p1, vector<PT> &p2) {
    double ans = inf;
    for (int i = 0; i < p1.size(); i++)
        ans = min(ans,
        dist_from_point_to_polygon(p2, p1[i]
        ));
    for (int i = 0; i < p2.size(); i++)
        ans = min(ans,
        dist_from_point_to_polygon(p1, p2[i]
        ));
    return ans;
}
// Maximum distance from a convex
// polygon to another convex polygon (O
// (n))
double
    maximum_dist_from_polygon_to_polygon(
    vector<PT> &u, vector<PT> &v) {
    int n = (int)u.size(), m = (int)v.size
    ();
    double ans = 0;
    if (n < 3 || m < 3) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) ans =
            max(ans, dist2(u[i], v[j]));
        }
        return sqrt(ans);
    }
    if (u[0].x > v[0].x) swap(n, m), swap(
    u, v);
    int i = 0, j = 0, step = n + m + 10;
    while (j + 1 < m && v[j].x < v[j + 1].
    x) j++;
    while (step-- > 0) {
        if (cross(u[(i + 1) % n] - u[i], v[(j
        + 1) % m] - v[j]) >= 0) j = (j + 1) %
        m;
        else i = (i + 1) % n;
        ans = max(ans, dist2(u[i], v[j]));
    }
    return sqrt(ans);
}
// BUGFIX: Using 'double' instead of '
// int' for coordinates to prevent
// precision loss.
double polygon_area(vector<PT> &v) {
    double calc_area = 0;
    int n = v.size();
    for (int i = 0; i < n; i++) {
        PT a = v[i], b = v[(i + 1) % n];
        double x1 = a.x, y1 = a.y; // Fixed
        from int
        double x2 = b.x, y2 = b.y; // Fixed
        from int
        calc_area += (x1 * y2 - x2 * y1);
    }
    return abs(calc_area) / 2.0;
}
// WARNING: It's highly recommended to
// use the standard '
// is_point_in_polygon()'
// above instead of this custom one,
// since 'seg_seg_intersection' returns
// a boolean
// making 'val == 1' or 'val == 2'
// evaluations buggy.
int position_of_a_point_in_a_polygon(
    vector<PT> &v, PT c) {
    double xInf = inf, yInf = c.y + 1;
    PT d(xInf, yInf);
    int n = v.size();
    int cnt = 0;
    for (int i = 0; i < n; i++) {
        PT a = v[i], b = v[(i + 1) % n];
        // This relies on your overridden
        // version of seg_seg_intersection
        // which returns bool.
        bool intersect =
        seg_seg_intersection(a, b, c, d);
        if (is_point_on_seg(a, b, c)) return
        0; // BOUNDARY
        if (intersect) cnt++;
    }
    if (cnt % 2) return 1; // INSIDE
    else return -1; // OUTSIDE
}

```

```

// 0 -> collinear, 1 -> clockwise, 2 ->
// counter-clockwise
int getSign(int x1, int y1, int x2, int
y2, int x3, int y3) {
    int direct = (y2 - y1) * (x3 - x2) - (
    y3 - y2) * (x2 - x1);
    if (direct > 0) return 1; //
    clockwise
    if (direct < 0) return 2; //
    counter - ClockWise
    return 0; //
    collinear
}
// 0 -> collinear, 1 -> clockwise, 2 ->
// counter-clockwise
int ori(int x1, int y1, int x2, int y2,
int x3, int y3) {
    int val = (y2 - y1) * (x3 - x2) - (x2
    - x1) * (y3 - y2);
    if (val == 0) return 0; //
    collinear
    else if (val < 0) return 2; //
    counter-clockwise
    return 1; //
    clockwise
}

```

11 Miscellaneous

11.1 Interval Container

```

set<pii>::iterator addInterval(set<pii> &
is, int L, int R) {
    if (L == R) return is.end();
    auto it = is.lower_bound({L, R}),
    before = it;
    while (it != is.end() && it->first <=
    R) {
        R = max(R, it->second);
        before = it = is.erase(it);
    }
    if (it != is.begin() && (--it)->second
    >= L) {
        L = min(L, it->first);
        R = max(R, it->second);
        is.erase(it);
    }
    return is.insert(before, {L, R});
}
void removeInterval(set<pii> &is, int L,
int R) {
    if (L == R) return;
    auto it = addInterval(is, L, R);
    auto r2 = it->second;
    if (it->first == L) is.erase(it);
    else (int&)it->second = L;
    if (R != r2) is.emplace(R, r2);
}

```

11.2 Efficient Mask Traversal

```

int mask = 0;
for (int curOp = 1; curOp < (1LL << n);
++curOp) {
    int curOp = __builtin_ctz(x);
    mask ^= (1LL << curOp);
}

```

11.3 Generationg Permutations

```

VI a;
//...
sort(all(a));
do {
    // Process permutation...
} while (next_permutation(all(a)));

```

11.4 Iterating over Submasks

```

int n = 17;
for (int mask = 0; mask < (1 << n); ++
mask) {
    for (int sub = mask; sub; sub = (sub -
    1) & mask) {
        // Process submask...
    }
}

```

```
}
}
```

11.5 Random

```
mt19937 rng(chrono::steady_clock::now().
    time_since_epoch().count());
int randint(int l, int r) {
    uniform_int_distribution<int> dist(l,
        r);
    return dist(rng);
}
```

11.6 Index compression

```
struct compress {
    vector<int> a;
    compress(vector<int>& v) : a(v) {
        sort(all(a));
        a.erase(unique(all(a)), a.end());
    }
    int size() { return a.size(); }
    int operator[](int val) { return
        lower_bound(all(a), val) - a.begin
        (); }
};
```

11.7 pbds

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <typename T>
using ordered_set = tree<T, null_type,
    less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
ordered_set<int> st;
// ordered multiset
ordered_set<pair<int, int>> multi;
// get index/ number of smaller elements
// than x
st.order_of_key(x);
// access by index
st.find_by_order(idx);
```

11.8 Bitset

```
bitset<N> b; // All 0s; b[i] access; b.
    test(i) bounds check
bitset<N> b(string("110")); // b[0]=0, b
    [1]=1, b[2]=1
b.set(i, v=1); b.reset(i); b.flip(i); //
    Single bit modify
b.set(); b.reset(); b.flip(); // All
    bits modify
b.count(); b.any(); b.all(); // popcount
    ; if 1+ bits set; if all bits set
b >= k; b <= k; b &= b2; b |= b2; b ^=
    b2; b = ~b; // O(N/64) bitwise
// GCC Specific (The good stuff):
b._Find_first(); // Index of first '1' (
    returns N if none)
b._Find_next(i); // Index of next '1'
    after i (returns N if none)
```

11.9 Primes

```
int BASES[] = {31, 37, 41, 43, 47, 53,
    59, 61, 67, 71, 137, 139, 149, 151,
    157, 257, 263, 269, 271, 277, 311,
    313, 317, 331, 347};
int MOD_32[] = {998244353, 999999937,
    1000000007, 1000000009, 1000000021,
    1000000103, 1000003931,
    1000004357, 2147483647};
11 MOD_1E12[] = {10000000000039,
    10000000000063, 10000000000091,
    10000000000121, 10000000000169,
    10000000000213, 10000000000223};
11 MOD_64[] = {100000000000000003,
    100000000000000009, 1000000000000000033,
    4611686018427387847, 9223372036854775783};

pair<int, int> ntt_primes = {{(119 <<
    23) + 1, 3}, {(5 << 25) + 1, 3},
```

```
{{(7 << 26) + 1, 3}, {(479 << 21) +
    1, 3}, {(483 << 21) + 5, 5}};
```

11.10 Digit Range Sum

```
auto qrsm = [](11 n) -> 11 {
    11 ans = 0;
    11 po = 1;
    11 nm = n;
    while (n) {
        11 tm = n / 10;
        11 rem = n % 10;
        11 s1 = rem * (rem - 1) / 2;
        11 past = (nm % po) + 111;
        ans += (tm * 4511) * po + (s1) * (po
            ) + (rem * past);
        po *= 10;
        n /= 10;
    }
    return ans;
};
auto rsm = [&](11 l, 11 r) -> 11 {
    return qrsm(r) - qrsm(l - 1);
};
```

11.11 Custom Hash

```
// Usage: unordered_set<pair<int, int>,
    pointhash> st;
struct pointhash {
    size_t operator()(const pair<int, int>
        & p) const {
        return hash<int>()(p.first) ^ (hash<
            int>()(p.second) << 1);
    }
};
struct custom_hash {
    static uint64_t splitmix64(uint64_t x)
    {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0
            xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0
            x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().
            time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};
unordered_map<long long, int,
    custom_hash> safe_map;
```

12 Formulae

13 Formulae

13.0.1 Inclusion-Exclusion Principle

Let S_j be the sum of sizes of all intersections of exactly j properties:

$$S_j = \sum_{1 \leq i_1 < \dots < i_j \leq n} |A_{i_1} \cap \dots \cap A_{i_j}|$$

- **Union (At least 1):**

$$|A_1 \cup \dots \cup A_n| = \sum_{j=1}^n (-1)^{j-1} S_j$$

- **Exactly k properties:**

$$E_k = \sum_{j=k}^n (-1)^{j-k} \binom{j}{k} S_j$$

- **At least k properties:**

$$L_k = \sum_{j=k}^n (-1)^{j-k} \binom{j-1}{k-1} S_j$$

Implementation Note:

If intersections depend only on the number of sets j , precompute $S_j = \binom{n}{j} \times \text{intersect}(j)$ in $O(n)$. Otherwise, use bitmask PIE in $O(2^n)$.

13.0.2 Extgcd and Diophantine

Linear Diophantine equation solves the equation $ax + by = c$ where $\gcd(a, b) \mid c$. Extended gcd calculates $ax + by = 1$ where $\gcd(a, b) = 1$. Suppose, $(x, y) = (x_0, y_0)$. Now we can write diophantine equation as $\frac{a}{g}x + \frac{b}{g}y = \frac{c}{g}$ where $g = \gcd(a, b)$. This equation can be put into extended gcd and we can obtain $(X, Y) = (\frac{c}{g}x_0, \frac{c}{g}y_0)$. Thus the general solution is obtained as:

$$x = X + \frac{b}{g}t \quad y = Y - \frac{a}{g}t \quad t \in \mathbb{Z}$$

Typically these type of problem requires a range of x, y ($x \in [\min x, \max x], y \in [\min y, \max y]$).

13.0.3 Combinatorics

1. $\sum_{0 \leq k \leq n} \binom{n-k}{k} = \text{Fib}_{n+1}$
2. $\binom{n}{k} = \binom{n}{n-k}$
3. $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
4. $k \binom{n}{k} = n \binom{n-1}{k-1}$
5. $\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$
6. $\sum_{i=0}^n \binom{n}{i} = 2^n$
7. $\sum_{i \geq 0} \binom{n}{2i} = 2^{n-1}$
8. $\sum_{i \geq 0} \binom{n}{2i+1} = 2^{n-1}$
9. $\sum_{i=0}^k (-1)^i \binom{n}{i} = (-1)^k \binom{n-1}{k}$
10. $\sum_{i=0}^k \binom{n+i}{i} = \sum_{i=0}^k \binom{n+i}{n} = \binom{n+k+1}{k}$
11. $1 \binom{n}{1} + 2 \binom{n}{2} + 3 \binom{n}{3} + \dots + n \binom{n}{n} = n 2^{n-1}$
12. $1^2 \binom{n}{1} + 2^2 \binom{n}{2} + 3^2 \binom{n}{3} + \dots + n^2 \binom{n}{n} = (n+1) 2^{n-2}$
13. **Vandermonde's Identify:**

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$
14. **Hockey-Stick Identify:**

$$n, r \in \mathbb{N}, n > r, \sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$
15. $\sum_{i=0}^k \binom{k}{i}^2 = \binom{2k}{k}$
16. $\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}$
17. $\sum_{k=q}^n \binom{n}{k} \binom{k}{q} = 2^{n-q} \binom{n}{q}$
18. $\sum_{i=0}^n k^i \binom{n}{i} = (k+1)^n$
19. $\sum_{i=0}^n \binom{2n}{i} = 2^{2n-1} + \frac{1}{2} \binom{2n}{n}$
20. $\sum_{i=1}^n \binom{n}{i} \binom{n-1}{i-1} = \binom{2n-1}{n-1}$
21. $\sum_{i=0}^n \binom{2n}{i}^2 = \frac{1}{2} \left(\binom{4n}{2n} + \binom{2n}{n}^2 \right)$

22. **Highest Power of 2 that divides $2^n C_n$:** Let x be the number of 1s in the binary representation. Then the number of odd terms will be 2^x . Let it form a sequence. The n -th value in the sequence (starting from $n = 0$) gives the highest power of 2 that divides $2^n C_n$.

23. **Pascal Triangle**

- (a) In a row p where p is a prime number, all the terms in that row except the 1s are multiples of p .
- (b) Parity: To count odd terms in row n , convert n to binary. Let x be the number of 1s in the binary representation. Then the number of odd terms will be 2^x .
- (c) Every entry in row $2^n - 1, n \geq 0$, is odd.

24. An integer $n \geq 2$ is prime if and only if all the intermediate binomial coefficients

$$\binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n-1} \text{ are divisible by } n.$$

25. **Kummer's Theorem:** For given integers $n \geq m \geq 0$ and a prime number p , the largest power of p dividing $\binom{n}{m}$ is equal to the number of carries when m is added to $n-m$ in base p . For implementation take inspiration from lucas theorem.

26. Number of different binary sequences of length n such that no two 0's are adjacent = Fib_{n+1}

27. **Combination with repetition:** Let's say we choose k elements from an n -element set, the order doesn't matter and each element can be chosen more than once. In that case, the number of different combinations is: $\binom{n+k-1}{k}$

28. Number of ways to divide n persons in $\frac{n}{k}$ equal groups i.e. each having size k is

$$\frac{n!}{k!^{\frac{n}{k}} \left(\frac{n}{k}\right)!} = \prod_{n \geq k} \binom{n-1}{k-1}$$

29. The number non-negative solution of the equation: $x_1 + x_2 + x_3 + \dots + x_k = n$ is $\binom{n+k-1}{n}$

30. Number of ways to choose n ids from 1 to b such that every id has distance at least $k = \binom{b-(n-1)(k-1)}{n}$

31. $\sum_{i=1,3,5,\dots}^{i \leq n} \binom{n}{i} a^{n-i} b^i = \frac{1}{2}((a+b)^n - (a-b)^n)$

32. $\sum_{i=0}^n \frac{\binom{n}{i}}{\binom{n}{i}} = \frac{\binom{n+1}{n}}{\binom{n}{n}}$

33. **Derangement:** a permutation of the elements of a set, such that no element appears in its original position. Let $d(n)$ be the number of derangements of the identity permutation of size n .

$$d(n) = (n-1)(d(n-1) + d(n-2)), \text{ where } d(0) = 1, d(1) = 0.$$

34. **Involutions:** permutations such that $p^2 = \text{identity permutation}$. $a_0 = a_1 = 1$ and $a_n = a_{n-1} + (n-1)a_{n-2}$ for $n > 1$.

35. Let $T(n, k)$ be the number of permutations of size n for which all cycles have length $\leq k$.

$$T(n, k) = \begin{cases} n! & ; n \leq k \\ n \cdot T(n-1, k) & \\ -F(n-1, k) & \\ \cdot T(n-k-1, k) & ; n > k \end{cases}$$

$$\text{Here } F(n, k) = n \cdot (n-1) \cdot \dots \cdot (n-k+1)$$

36. **Lucas Theorem**

- (a) If p is prime, then $\binom{p^a}{k} \equiv 0 \pmod{p}$
- (b) For non-negative integers m and n and a prime p , the following congruence relation holds: $\binom{m}{n} \equiv \prod_{i=0}^k \binom{m_i}{n_i} \pmod{p}$, where, $m = m_k p^k + m_{k-1} p^{k-1} + \dots + m_1 p + m_0$, and $n = n_k p^k + n_{k-1} p^{k-1} + \dots + n_1 p + n_0$ are the base p expansions of m and n respectively. This uses the convention that $\binom{m}{n} = 0$, when $m < n$.

$$\begin{aligned} 37. \sum_{i=0}^n \binom{n}{i} \cdot i^k &= \sum_{i=0}^n \binom{n}{i} \cdot \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot i^j \\ &= \sum_{i=0}^n \binom{n}{i} \cdot \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot j! \binom{n}{i} \\ &= \sum_{i=0}^n \frac{n!}{(n-i)!} \cdot \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot \frac{1}{(i-j)!} \\ &= \sum_{i=0}^n \sum_{j=0}^k \frac{n!}{(n-i)!} \cdot \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot \frac{1}{(i-j)!} \\ &= n! \sum_{i=0}^n \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot \frac{1}{(n-i)!} \cdot \frac{1}{(i-j)!} \\ &= n! \sum_{i=0}^n \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot \binom{n-j}{n-i} \cdot \frac{1}{(n-j)!} \\ &= n! \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot \frac{1}{(n-j)!} \sum_{i=0}^n \binom{n-j}{n-i} \\ &= \sum_{j=0}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \cdot n^j \cdot 2^{n-j} \end{aligned}$$

Here $n^j = P(n, j) = \frac{n!}{(n-j)!}$ and $\left\{ \begin{matrix} k \\ j \end{matrix} \right\}$ is stirling number of the second kind.

So, instead of $O(n)$, now you can calculate the original equation in $O(k^2)$ or even in $O(k \log^2 n)$ using NTT.

$$\begin{aligned} 38. \sum_{i=0}^{n-1} \binom{i}{j} x^i &= x^j (1-x)^{-j-1} \\ &\cdot \left(1 - x^n \sum_{i=0}^j \binom{n}{i} x^{j-i} (1-x)^i \right) \end{aligned}$$

$$39. x_0, x_1, x_2, x_3, \dots, x_n$$

$$x_0 + x_1, x_1 + x_2, x_2 + x_3, \dots, x_{n-1} + x_n$$

...

If we continuously do this n times then the polynomial of the first column of the n -th row will be

$$p(n) = \sum_{k=0}^n \binom{n}{k} \cdot x(k)$$

$$40. \text{ If } P(n) = \sum_{k=0}^n \binom{n}{k} \cdot Q(k), \text{ then,}$$

$$Q(n) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} \cdot P(k)$$

$$41. \text{ If } P(n) = \sum_{k=0}^n (-1)^k \binom{n}{k} \cdot Q(k), \text{ then,}$$

$$Q(n) = \sum_{k=0}^n (-1)^k \binom{n}{k} \cdot P(k)$$

13.0.4 **Catalan Numbers**

$$42. C_n = \frac{1}{n+1} \binom{2n}{n}$$

$$43. C_0 = 1, C_1 = 1 \text{ and } C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$$

44. Number of correct bracket sequence consisting of n opening and n closing brackets.

45. The number of ways to completely parenthesize $n+1$ factors.

46. The number of triangulations of a convex polygon with $n+2$ sides (i.e. the number of partitions of polygon into disjoint triangles by using the diagonals).

47. The number of ways to connect the $2n$ points on a circle to form n disjoint i.e. non-intersecting chords.

48. The number of monotonic lattice paths from point $(0,0)$ to point (n,n) in a square lattice of size $n \times n$, which do not pass above the main diagonal (i.e. connecting $(0,0)$ to (n,n)).

49. The number of rooted full binary trees with $n+1$ leaves (vertices are not numbered). A rooted binary tree is full if every vertex has either two children or no children.

50. Number of permutations of $\{1, \dots, n\}$ that avoid the pattern 123 (or any of the other patterns of length 3); that is, the number of permutations with no three-term increasing subsequence. For $n = 3$, these permutations are 132, 213, 231, 312 and 321. For $n = 4$, they are 1432, 2143, 2413, 2431, 3142, 3214, 3241, 3412, 3421, 4132, 4213, 4231, 4312 and 4321.

51. **Balanced Parentheses count with prefix:** The count of balanced parentheses sequences consisting of $n+k$ pairs of parentheses where the first k symbols are open brackets. Let the number be $C_n^{(k)}$, then

$$C_n^{(k)} = \frac{k+1}{n+k+1} \binom{2n+k}{n}$$

13.0.5 **Narayana numbers**

$$52. N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}$$

53. The number of expressions containing n pairs of parentheses, which are correctly matched and which contain k distinct nestings. For instance, $N(4, 2) = 6$ as with four pairs of parentheses six sequences can be created which each contain two times the sub-pattern '()'.

13.0.6 **Stirling numbers of the first kind**

54. The Stirling numbers of the first kind count permutations according to their number of cycles (counting fixed points as cycles of length one).

55. $S(n, k)$ counts the number of permutations of n elements with k disjoint cycles.

56. $S(n, k) = (n-1) \cdot S(n-1, k) + S(n-1, k-1)$, where, $S(0, 0) = 1, S(n, 0) = S(0, n) = 0$

$$57. \sum_{k=0}^n S(n, k) = n!$$

58. The unsigned Stirling numbers may also be defined algebraically, as the coefficient of the rising factorial:

$$x^{\bar{n}} = x(x+1) \dots (x+n-1) = \sum_{k=0}^n S(n, k) x^k$$

59. Lets $[n, k]$ be the stirling number of the first kind, then

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \sum_{0 \leq i_1 < i_2 < \dots < i_k < n} i_1 i_2 \dots i_k$$

13.0.7 Stirling numbers of the second kind

60. Stirling number of the second kind is the number of ways to partition a set of n objects into k non-empty subsets.
61. $S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$, where $S(0, 0) = 1, S(n, 0) = S(0, n) = 0$
62. $S(n, 2) = 2^{n-1} - 1$
63. $S(n, k) \cdot k!$ = number of ways to color n nodes using colors from 1 to k such that each color is used at least once.
64. An r -associated Stirling number of the second kind is the number of ways to partition a set of n objects into k subsets, with each subset containing at least r elements. It is denoted by $S_r(n, k)$ and obeys the recurrence relation. $S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1)$
65. Denote the n objects to partition by the integers $1, 2, \dots, n$. Define the reduced Stirling numbers of the second kind, denoted $S^d(n, k)$, to be the number of ways to partition the integers $1, 2, \dots, n$ into k nonempty subsets such that all elements in each subset have pairwise distance at least d . That is, for any integers i and j in a given subset, it is required that $|i-j| \geq d$. It has been shown that these numbers satisfy, $S^d(n, k) = S(n-d+1, k-d+1), n \geq k \geq d$

13.0.8 Bell number

66. Counts the number of partitions of a set.
67. $B_{n+1} = \sum_{k=0}^n \binom{n}{k} \cdot B_k$
68. $B_n = \sum_{k=0}^n S(n, k)$, where $S(n, k)$ is Stirling number of second kind.

13.0.9 Math

69. $ab \bmod ac = a(b \bmod c)$
70. $\sum_{i=0}^n i \cdot i! = (n+1)! - 1$.
71. $a^k - b^k = (a-b) \cdot (a^{k-1}b^0 + a^{k-2}b^1 + \dots + a^0b^{k-1})$
72. $\min(a+b, c) = a + \min(b, c-a)$
73. $|a-b| + |b-c| + |c-a| = 2(\max(a, b, c) - \min(a, b, c))$
74. $a \cdot b \leq c \rightarrow a \leq \left\lfloor \frac{c}{b} \right\rfloor$ is correct
75. $a \cdot b < c \rightarrow a < \left\lfloor \frac{c}{b} \right\rfloor$ is incorrect
76. $a \cdot b \geq c \rightarrow a \geq \left\lfloor \frac{c}{b} \right\rfloor$ is correct
77. $a \cdot b > c \rightarrow a > \left\lfloor \frac{c}{b} \right\rfloor$ is correct
78. For positive integer n , and arbitrary real numbers m, x , $\left\lfloor \frac{\lfloor x/m \rfloor}{n} \right\rfloor = \left\lfloor \frac{x}{mn} \right\rfloor$
79. Lagrange's identity:

$$\left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) - \left(\sum_{k=1}^n a_k b_k \right)^2 = \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_i b_j - a_j b_i)^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1, j \neq i}^n (a_i b_j - a_j b_i)^2$$
80. $\sum_{i=1}^n i a^i = \frac{a(na^{n+1} - (n+1)a^n + 1)}{(a-1)^2}$

81. Vieta's formulas: Any general polynomial of degree n

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

(with the coefficients being real or complex numbers and $a_n \neq 0$) is known by the fundamental theorem of algebra to have n (not necessarily distinct) complex roots $r_1, r_2, \dots, r_n$.

$$\begin{cases} r_1 + r_2 + \dots + r_{n-1} + r_n = -\frac{a_{n-1}}{a_n} \\ (r_1 r_2 + r_1 r_3 + \dots + r_1 r_n) \\ + (r_2 r_3 + r_2 r_4 + \dots + r_2 r_n) + \dots \\ + r_{n-1} r_n = \frac{a_{n-2}}{a_n} \\ \vdots \\ r_1 r_2 \dots r_n = (-1)^n \frac{a_0}{a_n} \end{cases}$$

Vieta's formulas can equivalently be written as

$$\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \left(\prod_{j=1}^k r_{i_j} \right) = (-1)^k \frac{a_{n-k}}{a_n}$$

82. We are given n numbers $a_1, a_2, \dots, a_n$ and our task is to find a value x that minimizes the sum,

$$|a_1 - x| + |a_2 - x| + \dots + |a_n - x|$$

optimal x = median of the array. if n is even $x = [\text{left median}, \text{right median}]$ i.e. every number in this range will work. For minimizing

$$(a_1 - x)^2 + (a_2 - x)^2 + \dots + (a_n - x)^2$$

$$\text{optimal } x = \frac{(a_1 + a_2 + \dots + a_n)}{n}$$

83. Given an array a of n non-negative integers. The task is to find the sum of the product of elements of all the possible subsets. It is equal to the product of $(a_i + 1)$ for all a_i

84. Pentagonal number theorem: In mathematics, the pentagonal number theorem states that

$$\prod_{n=1}^{\infty} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{\frac{k(3k-1)}{2}} = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{\frac{k(3k+1)}{2}} + x^{\frac{k(3k-1)}{2}} \right)$$

In other words,

$$(1-x)(1-x^2)(1-x^3) \dots = 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + x^{22} + x^{26} - \dots$$

The exponents $1, 2, 5, 7, 12, \dots$ on the right hand side are given by the formula $g_k = \frac{k(3k-1)}{2}$ for $k = 1, -1, 2, -2, 3, \dots$ and are called (generalized) pentagonal numbers. It is useful to find the partition number in $O(n\sqrt{n})$

13.0.10 Fibonacci Number

$$85. F_0 = 0, F_1 = 1 \text{ and } F_n = F_{n-1} + F_{n-2}$$

$$86. F_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n-k-1}{k}$$

$$87. F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$88. \sum_{i=1}^n F_i = F_{n+2} - 1$$

$$89. \sum_{i=0}^{n-1} F_{2i+1} = F_{2n}$$

$$90. \sum_{i=1}^n F_{2i} = F_{2n+1} - 1$$

$$91. \sum_{i=1}^n F_i^2 = F_n F_{n+1}$$

$$92. F_m F_{n+1} - F_{m-1} F_n = (-1)^n F_{m-n}$$

$$93. F_m F_n + F_{m-1} F_{n-1} = F_{m+n-1}$$

94. A number is Fibonacci if and only if one or both of $(5 \cdot n^2 + 4)$ or $(5 \cdot n^2 - 4)$ is a perfect square

95. Every third number of the sequence is even and more generally, every k^{th} number of the sequence is a multiple of F_k

$$96. \gcd(F_m, F_n) = F_{\gcd(m, n)}$$

97. Any three consecutive Fibonacci numbers are pairwise coprime, which means that, for every n , $\gcd(F_n, F_{n+1}) = \gcd(F_n, F_{n+2}), \gcd(F_{n+1}, F_{n+2}) = 1$

98. If the members of the Fibonacci sequence are taken *mod* n , the resulting sequence is periodic with period at most $6n$.

13.0.11 Pythagorean Triples

99. A Pythagorean triple consists of three positive integers a, b , and C , such that $a^2 + b^2 = c^2$. Such a triple is commonly written (a, b, c)

100. Euclid's formula is a fundamental formula for generating Pythagorean triples given an arbitrary pair of integers m and n with $m > n > 0$. The formula states that the integers

$$a = m^2 - n^2, b = 2mn, c = m^2 + n^2$$

form a Pythagorean triple. The triple generated by Euclid's formula is primitive if and only if m and n are coprime and not both odd. When both m and n are odd, then a, b , and c will be even, and the triple will not be primitive; however, dividing a, b , and c by 2 will yield a primitive triple when m and n are coprime and both odd.

101. The following will generate all Pythagorean triples uniquely:

$$a = k(m^2 - n^2), b = k(2mn), c = k(m^2 + n^2).$$

where m, n , and k are positive integers with $m > n$, and with m and n coprime and not both odd.

102. **Theorem:** The number of Pythagorean triples a, b, n with $\max(a, b, n) = n$ is given by

$$\frac{1}{2} \left(\prod_{p^\alpha || n} (2\alpha + 1) - 1 \right)$$

where the product is over all prime divisors p of the form $4k+1$. The notation $p^\alpha || n$ stands for the highest exponent α for which p^α divides n . **Example:** For $n = 2 \cdot 3^2 \cdot 5^3 \cdot 7^4 \cdot 11^5 \cdot 13^6$, the number of Pythagorean triples with hypotenuse n is $\frac{1}{2} (7 \cdot 13 - 1) = 45$. To obtain a formula for the number of Pythagorean triples with hypotenuse less than a specific positive integer N , we may add the numbers corresponding to each $n < N$ given by the Theorem. There is no simple way to compute this as a function of N .

13.0.12 Sum of Squares Function

103. The function is defined as

$$r_k(n) = \left| \left\{ (a_1, a_2, \dots, a_k) \in \mathbf{Z}^k : n = a_1^2 + a_2^2 + \dots + a_k^2 \right\} \right|$$

104. The number of ways to write a natural number as sum of two squares is given by $r_2(n)$. It is given explicitly by $r_2(n) = 4(d_1(n) - d_3(n))$ where $d_1(n)$ is the number of divisors of n which are congruent with 1 modulo 4 and $d_3(n)$ is the

number of divisors of n which are congruent with 3 modulo 4. The prime factorization $n = 2^g p_1^{f_1} p_2^{f_2} \dots q_1^{h_1} q_2^{h_2} \dots$, where p_i are the prime factors of the form $p_i \equiv 1 \pmod{4}$, and q_i are the prime factors of the form $q_i \equiv 3 \pmod{4}$ gives another formula $r_2(n) = 4(f_1 + 1)(f_2 + 1) \dots$, if all exponents $h_1, h_2, \dots$ are even. If one or more h_i are odd, then $r_2(n) = 0$.

105. The number of ways to represent n as the sum of four squares is eight times the sum of all its divisors which are not divisible by 4, i.e. $r_4(n) = 8 \sum_{d|n; 4 \nmid d} d$

$$r_8(n) = 16 \sum_{d|n} (-1)^{n+d} d^3$$

13.0.13 Number Theory

106. for $i > j$, $\gcd(i, j) = \gcd(i - j, j) \leq (i - j)$

$$107. \sum_{x=1}^n [dx^k] = \left\lfloor \frac{n}{\prod_{i=0}^k p_i^{\lceil \frac{e_i}{k} \rceil}} \right\rfloor, \text{ where } d = \prod_{i=0}^k p_i^{e_i}.$$

Here, $[a/b]$ means if a divides b then it is 1, otherwise it is 0.

108. The number of lattice points on segment (x_1, y_1) to (x_2, y_2) is $\gcd(\text{abs}(x_1 - x_2), \text{abs}(y_1 - y_2)) + 1$
109. $(n - 1)! \pmod n = n - 1$ if n is prime, 2 if $n = 4$, 0 otherwise.
110. A number has odd number of divisors if it is perfect square
111. The sum of all divisors of a natural number n is odd if and only if $n = 2^r \cdot k^2$ where r is non-negative and k is positive integer.

112. Let a and b be coprime positive integers, and find integers a' and b' such that $aa' \equiv 1 \pmod b$ and $bb' \equiv 1 \pmod a$. Then the number of representations of a positive integers (n) as a non negative linear combination of a and b is

$$\frac{n}{ab} - \left\{ \frac{bn}{a} \right\} - \left\{ \frac{an}{b} \right\} + 1$$

Here, x denotes the fractional part of x .

$$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c d(i \cdot j \cdot k)$$

$$= \sum_{\gcd(i,j)=\gcd(j,k)=\gcd(k,i)=1} \left\lfloor \frac{a}{i} \right\rfloor \left\lfloor \frac{b}{j} \right\rfloor \left\lfloor \frac{c}{k} \right\rfloor$$

Here, $d(x)$ = number of divisors of x .

113. **Gauss's generalization of Wilson's theorem:** Gauss proved that,

$$\prod_{\substack{k=1 \\ \gcd(k,m)=1}}^m k \equiv \begin{cases} -1 \pmod m & \text{if } m = 4, p^\alpha, 2p^\alpha \\ 1 \pmod m & \text{otherwise} \end{cases}$$

where p represents an odd prime and α a positive integer. The values of m for which the product is -1 are precisely the ones where there is a primitive root modulo m .

13.0.14 Divisor Function

$$114. \sigma_x(n) = \sum_{d|n} d^x$$

115. It is multiplicative i.e if $\gcd(a, b) = 1 \rightarrow \sigma_x(ab) = \sigma_x(a)\sigma_x(b)$.

$$116. \sigma_x(n) = \prod_{i=1}^{\tau} \frac{p_i^{(a_i+1)x} - 1}{p_i^x - 1}$$

117. Divisor Summatory Function

- (a) Let $\sigma_0(k)$ be the number of divisors of k .

$$(b) D(x) = \sum_{n \leq x} \sigma_0(n)$$

$$(c) D(x) = \sum_{k=1}^x \left\lfloor \frac{x}{k} \right\rfloor = 2 \sum_{k=1}^u \left\lfloor \frac{x}{k} \right\rfloor - u^2, \text{ where } u = \sqrt{x}$$

- (d) $D(n)$ = Number of increasing arithmetic progressions where $n + 1$ is the second or later term. (i.e. The last term, starting term can be any positive integer $\leq n$. For example, $D(3) = 5$ and there are 5 such arithmetic progressions: $(1, 2, 3, 4); (2, 3, 4); (1, 4); (2, 4); (3, 4)$).

118. Let $\sigma_1(k)$ be the sum of divisors of k .

$$\text{Then, } \sum_{k=1}^n \sigma_1(k) = \sum_{k=1}^n k \left\lfloor \frac{n}{k} \right\rfloor$$

119. $\prod_{d|n} d = n^{\frac{\sigma_0(n)}{2}}$ if n is not a perfect square, and $= \sqrt{n} \cdot n^{\frac{\sigma_0(n)-1}{2}}$ if n is a perfect square.

13.0.15 Euler's Totient function

120. The function is multiplicative. This means that if $\gcd(m, n) = 1$, $\phi(m \cdot n) = \phi(m) \cdot \phi(n)$.

$$121. \phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

122. If p is prime and $(k \geq 1)$, then: $\phi(p^k) = p^{k-1}(p - 1) = p^k \left(1 - \frac{1}{p}\right)$

123. $J_k(n)$, the Jordan totient function, is the number of k -tuples of positive integers all less than or equal to n that form a coprime $(k + 1)$ -tuple together with n . It is a generalization of Euler's totient, $\phi(n) = J_1(n)$. $J_k(n) = n^k \prod_{p|n} \left(1 - \frac{1}{p^k}\right)$

$$124. \sum_{d|n} J_k(d) = n^k$$

$$125. \sum_{d|n} \phi(d) = n$$

$$126. \phi(n) = \sum_{d|n} \mu(d) \cdot \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d}$$

$$127. \phi(n) = \sum_{d|n} d \cdot \mu\left(\frac{n}{d}\right)$$

$$128. a|b \rightarrow \varphi(a)|\varphi(b)$$

$$129. n|\varphi(a^n - 1) \text{ for } a, n > 1$$

$$130. \varphi(mn) = \varphi(m)\varphi(n) \cdot \frac{d}{\varphi(d)} \text{ where } d = \gcd(m, n) \text{ Note the special cases}$$

$$\varphi(2m) = \begin{cases} 2\varphi(m) & \text{if } m \text{ is even} \\ \varphi(m) & \text{if } m \text{ is odd} \end{cases}$$

$$\varphi(n^m) = n^{m-1}\varphi(n)$$

131. $\varphi(\text{lcm}(m, n)) \cdot \varphi(\gcd(m, n)) = \varphi(m) \cdot \varphi(n)$
Compare this to the formula $\text{lcm}(m, n) \cdot \gcd(m, n) = m \cdot n$

132. $\varphi(n)$ is even for $n \geq 3$. Moreover, if n has r distinct odd prime factors, $2^r | \varphi(n)$

$$133. \sum_{d|n} \frac{\mu^2(d)}{\varphi(d)} = \frac{n}{\varphi(n)}$$

$$134. \sum_{1 \leq k \leq n, \gcd(k, n)=1} k = \frac{1}{2} n \varphi(n) \text{ for } n > 1$$

$$135. \frac{\varphi(n)}{n} = \frac{\varphi(\text{rad}(n))}{\text{rad}(n)} \text{ where } \text{rad}(n) = \prod_{p|n, p \text{ prime}} p$$

$$136. \phi(m) \geq \log_2 m$$

$$137. \phi(\phi(m)) \leq \frac{m}{2}$$

138. When $x \geq \log_2 m$, then

$$n^x \pmod m = n^{\phi(m)+x \pmod{\phi(m)}} \pmod m$$

139. $\sum_{1 \leq k \leq n, \gcd(k, n)=1} \gcd(k - 1, n) = \varphi(n)d(n)$ where $d(n)$ is number of divisors. Same equation for $\gcd(a \cdot k - 1, n)$ where a and n are coprime.

140. For every n there is at least one other integer $m \neq n$ such that $\varphi(m) = \varphi(n)$.

$$141. \sum_{i=1}^n \varphi(i) \cdot \left\lfloor \frac{n}{i} \right\rfloor = \frac{n \cdot (n+1)}{2}$$

142. $\sum_{i=1, i \% 2 \neq 0}^n \varphi(i) \cdot \left\lfloor \frac{n}{i} \right\rfloor = \sum_{k \geq 1} \left\lfloor \frac{n}{2k} \right\rfloor^2$. Note that $[]$ is used here to denote round operator not floor or ceil

- 143.

$$\sum_{i=1}^n \sum_{j=1}^n ij [\gcd(i, j) = 1] = \sum_{i=1}^n \varphi(i) i^2$$

144. Average of coprimes of n which are less than n is $\frac{n}{2}$.

13.0.16 Mobius Function and Inversion

145. It is a multiplicative function.

- 146.

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & ; n = 1 \\ 0 & ; n > 0 \end{cases}$$

147. $\sum_{n=1}^N \mu^2(n) = \sum_{n=1}^{\sqrt{N}} \mu(k) \cdot \left\lfloor \frac{N}{k^2} \right\rfloor$ This is also the number of square-free numbers $\leq n$

148. **Mobius inversion theorem:** The classic version states that if g and f are arithmetic functions satisfying $g(n) = \sum_{d|n} f(d)$ for every integer $n \geq 1$ then $f(n) = \sum_{d|n} \mu(d)g\left(\frac{n}{d}\right)$ for every integer $n \geq 1$

$$149. \text{ If } F(n) = \prod_{d|n} f(d), \text{ then } F(n) = \prod_{d|n} F\left(\frac{n}{d}\right)^{\mu(d)}$$

150. $\sum_{d|n} \mu(d)\phi(d) = \prod_{j=1}^K (2 - p_j)$ where p_j is the primes factorization of d

151. If $F(n)$ is multiplicative, $F \neq 0$, then $\sum_{d|n} \mu(d)f(d) = \prod_{i=1}^r (1 - f(p_i))$ where p_i are primes of n .

13.0.17 GCD and LCM

152. $\gcd(a, 0) = a$
153. $\gcd(a, b) = \gcd(b, a \bmod b)$
154. Every common divisor of a and b is a divisor of $\gcd(a, b)$.
155. if m is any integer, then $\gcd(a + m \cdot b, b) = \gcd(a, b)$
156. The gcd is a multiplicative function in the following sense: if a_1 and a_2 are relatively prime, then $\gcd(a_1 \cdot a_2, b) = \gcd(a_1, b) \cdot \gcd(a_2, b)$.
157. $\gcd(a, b) \cdot \text{lcm}(a, b) = |a \cdot b|$
158. $\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$.
159. $\text{lcm}(a, \gcd(b, c)) = \gcd(\text{lcm}(a, b), \text{lcm}(a, c))$.
160. For non-negative integers a and b , where a and b are not both zero, $\gcd(n^a - 1, n^b - 1) = n^{\gcd(a, b)} - 1$
161. $\gcd(a, b) = \sum_{k|a \text{ and } k|b} \phi(k)$
162. $\sum_{i=1}^n [\gcd(i, n) = k] = \phi\left(\frac{n}{k}\right)$
163. $\sum_{k=1}^n \gcd(k, n) = \sum_{d|n} d \cdot \phi\left(\frac{n}{d}\right)$
164. $\sum_{k=1}^n x^{\gcd(k, n)} = \sum_{d|n} x^d \cdot \phi\left(\frac{n}{d}\right)$
165. $\sum_{k=1}^n \frac{1}{\gcd(k, n)} = \sum_{d|n} \frac{1}{d} \cdot \phi\left(\frac{n}{d}\right) = \frac{1}{n} \sum_{d|n} d \cdot \phi(d)$
166. $\sum_{k=1}^n \frac{k}{\gcd(k, n)} = \frac{n}{2} \cdot \sum_{d|n} \frac{1}{d} \cdot \phi\left(\frac{n}{d}\right) = \frac{n}{2} \cdot \frac{1}{n} \cdot \sum_{d|n} d \cdot \phi(d)$
167. $\sum_{k=1}^n \frac{n}{\gcd(k, n)} = 2 * \sum_{k=1}^n \frac{k}{\gcd(k, n)} - 1$, for $n > 1$
168. $\sum_{i=1}^n \sum_{j=1}^n [\gcd(i, j) = 1] = \sum_{d=1}^n \mu(d) \left\lfloor \frac{n}{d} \right\rfloor^2$
169. $\sum_{i=1}^n \sum_{j=1}^n \gcd(i, j) = \sum_{d=1}^n \phi(d) \left\lfloor \frac{n}{d} \right\rfloor^2$
170. $\sum_{i=1}^n \sum_{j=1}^n i \cdot j [\gcd(i, j) = 1] = \sum_{i=1}^n \phi(i) i^2$
171. $F(n) = \sum_{i=1}^n \sum_{j=1}^n \text{lcm}(i, j) = \sum_{l=1}^n \left(\frac{(1 + \lfloor \frac{n}{l} \rfloor) (\lfloor \frac{n}{l} \rfloor)}{2} \right) \sum_{d|l} \mu(d) d$
172. $\gcd(\text{lcm}(a, b), \text{lcm}(b, c), \text{lcm}(a, c)) = \text{lcm}(\gcd(a, b), \gcd(b, c), \gcd(a, c))$
173. $\gcd(A_L, A_{L+1}, \dots, A_R) = \gcd(A_L, A_{L+1} - A_L, \dots, A_R - A_{R-1})$.
174. Given n , If $SUM = LCM(1, n) + LCM(2, n) + \dots + LCM(n, n)$ then $SUM = \frac{n}{2} \left(\sum_{d|n} (\phi(d) \times d) + 1 \right)$

13.0.18 Legendre Symbol

175. $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$ and $\left(\frac{a}{p}\right) \in \{-1, 0, 1\}$.
176. The Legendre symbol is periodic in its first (or top) argument: if $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.

177. The Legendre symbol is a completely multiplicative function of its top argument:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$$
178. The Fibonacci numbers $1, 1, 2, 3, 5, 8, 13, \dots$ are defined by the recurrence $F_1 = F_2 = 1, F_{n+1} = F_n + F_{n-1}$. If p is a prime number then

$$F_{p - (\frac{p}{5})} \equiv 0 \pmod{p}, \quad F_p \equiv \left(\frac{p}{5}\right) \pmod{p}.$$
179. Continuing from previous point,
 $\left(\frac{p}{5}\right) =$ infinite concatenation of the sequence $(1, -1, -1, 1, 0)$ from $p \geq 1$.
180. If $n = k^2$ is perfect square then $\left(\frac{n}{p}\right) = 1$ for every odd prime except $\left(\frac{n}{k}\right) = 0$ if k is an odd prime.

13.0.19 Miscellaneous

181. $a + b = a \oplus b + 2(a \& b)$.
182. $a + b = a | b + a \& b$
183. $a \oplus b = a | b - a \& b$
184. k_{th} bit is set in x iff $x \bmod 2^{k-1} \geq 2^k$. It comes handy when you need to look at the bits of the numbers which are pair sums or subset sums etc.
185. k_{th} bit is set in x iff $x \bmod 2^{k-1} - x \bmod 2^k \neq 0$ ($= 2^k$ to be exact). It comes handy when you need to look at the bits of the numbers which are pair sums or subset sums etc.
186. $n \bmod 2^i = n \& (2^i - 1)$
187. $1 \oplus 2 \oplus 3 \oplus \dots \oplus (4k - 1) = 0$ for any $k \geq 0$
188. **Erdos Gallai Theorem:** The degree sequence of an undirected graph is the non-increasing sequence of its vertex degrees. A sequence of non-negative integers $d_1 \geq d_2 \geq \dots \geq d_n$ can be represented as the degree sequence of finite simple graph on n vertices if and only if $d_1 + d_2 + \dots + d_n$ is even and

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$
holds for every k in $1 \leq k \leq n$.
189. $f_n = \sum_{i=0}^n \binom{n}{i} g_i \Rightarrow g_n = \sum_{i=0}^n (-1)^{n-i} \binom{n}{i} f_i$

13.0.20 Traingle

- Sine Rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$
- Cosine Rule:** $c^2 = a^2 + b^2 - 2ab \cos C$
- Area:** $\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \frac{1}{2}bc \sin A$
- Circumradius (R):** $R = \frac{abc}{4\Delta} = \frac{a}{2 \sin A}$
- Inradius (r):** $r = \frac{\Delta}{s}$
- Ex-radii (r_a, r_b, r_c):** $r_a = \frac{\Delta}{s-a}$
- Centroid (G):** $\left(\frac{\sum x}{3}, \frac{\sum y}{3}\right)$
- Incenter (I):** $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$
- Excenter (I_A):** Replace a with $-a$ in Incenter formula.
- Euler Line:** O, G, H are collinear; $HG = 2GO$.
- Median (m_a):** $4m_a^2 = 2b^2 + 2c^2 - a^2$ (Apollonius)
- Angle Bisector (l_a):** $l_a = \frac{2bc}{b+c} \cos(A/2)$

- Stewart's Theorem:** Cevian (a line from A to a) length d dividing a into m, n :

$$b^2m + c^2n = a(d^2 + mn)$$

- Centroid (G):** Intersection of *medians* (vertex to midpoint).
- Incenter (I):** Intersection of *angle bisectors*.
- Circumcenter (O):** Intersection of *perpendicular bisectors* of the sides.
- Orthocenter (H):** Intersection of *altitudes* (vertex perpendicular to opposite side).

13.0.21 Circle

- General Eq:** $x^2 + y^2 + 2gx + 2fy + c = 0$. Center $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$.
- Tangent at $P(x_1, y_1)$:** $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- Slope of tangent from $P(x, y)$:** $y + f = m(x + g) \pm r\sqrt{1 + m^2}$
- Tangency Condition ($y = mx + c$):** $c^2 = r^2(1 + m^2)$
- Chord Length:** $L = 2\sqrt{r^2 - d^2}$ (d = dist to center)
- Power of Point P :** $d^2 - r^2$ (d = dist P to center) (Useful for determining if a point is inside or outside)
- Secant Theorem:** $PA \cdot PB = PC \cdot PD$ (AB and CD are 2 lines drawn from point P over the circle)
- Sector Area:** $\frac{1}{2}r^2\theta$ (θ in rads)
- Segment Area (Cut by chord):** $\frac{1}{2}r^2(\theta - \sin \theta)$

13.0.22 Points & 2D Line

- Dist. point (x_1, y_1) to line $ax + by + c = 0$:**

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$
- Dist. between parallel lines:** $d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$
- Shoelace Area:** $\frac{1}{2} |\sum (x_i y_{i+1} - x_{i+1} y_i)|$
- foot of Perpendicular (h, k) :**

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$$
- Reflection (h, k) :** Replace $-(...)$ with $-2(...)$ above.
- Rotation of Axes (CCW θ):** $x = X \cos \theta - Y \sin \theta$, $y = X \sin \theta + Y \cos \theta$

13.0.23 Vector

- Direction Cosines:** $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$. $l^2 + m^2 + n^2 = 1$.
- Dot Product:** $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$
- Projection of a on b:** $(\mathbf{a} \cdot \hat{\mathbf{b}}) \hat{\mathbf{b}}$
- Cross Product:** Area $\Delta = \frac{1}{2} |\mathbf{a} \times \mathbf{b}|$.
- Scalar Triple (Vol):** $[\mathbf{abc}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. Tet Vol = $\frac{1}{6} |[\dots]|$.
- Vector Triple:** $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}$

13.0.24 3D Line

- Line 1: $\mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1$. Line 2: $\mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2$.
- Angle Between Lines:** $\cos \theta = \frac{|\mathbf{b}_1 \cdot \mathbf{b}_2|}{|\mathbf{b}_1| |\mathbf{b}_2|}$
 - Condition for Intersection:** Scalar Triple Product $[\mathbf{a}_2 - \mathbf{a}_1, \mathbf{b}_1, \mathbf{b}_2] = 0$.
 - Intersection Point:** Solve $\mathbf{a}_1 + \lambda \mathbf{b}_1 = \mathbf{a}_2 + \mu \mathbf{b}_2$ for λ, μ .

- **Shortest Dist (Skew):** $D = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2)|}{|\mathbf{b}_1 \times \mathbf{b}_2|}$
- **Dist Between Parallel Lines:** $D = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \times \mathbf{b}|}{|\mathbf{b}|}$
- **Dist Point P to Line:** $D = \frac{|(\mathbf{p} - \mathbf{a}) \times \mathbf{b}|}{|\mathbf{b}|}$
- **Foot of Perp F from P :** $\mathbf{f} = \mathbf{a} + \left(\frac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b}$
- **Image P' in Line:** $\mathbf{p}' = 2\mathbf{f} - \mathbf{p}$.

13.0.25 Plane

Through $\mathbf{a}$, normal $\mathbf{n}$. Eq: $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.

- **Cartesian:** $Ax + By + Cz + D = 0$. Normal (A, B, C) .
- **Dist Point P to Plane:** $D = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$
- **Angle Between Planes:** $\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$
- **Angle Between Line & Plane:** $\sin \phi = \frac{|\mathbf{b} \cdot \mathbf{n}|}{|\mathbf{b}||\mathbf{n}|}$
- **Intersection (Line of two planes):** Direction $\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2$.
- **Family of Planes:** $P_1 + \lambda P_2 = 0$ (Planes through intersection line).
- **Foot of Perp from P :** $\frac{x - x_1}{A} = \frac{y - y_1}{B} = \frac{z - z_1}{C} = \frac{-(Ax_1 + By_1 + Cz_1 + D)}{A^2 + B^2 + C^2}$
- **Image of Point in Plane:** Use -2 instead of -1 above.
- **Plane Bisectors:** $\frac{A_1x + B_1y + C_1z + D_1}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = \pm \frac{A_2x + B_2y + C_2z + D_2}{\sqrt{A_2^2 + B_2^2 + C_2^2}}$

13.0.26 Sphere

- **Standard:** $(x - u)^2 + (y - v)^2 + (z - w)^2 = r^2$.
- **Diameter Form:** $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$.
- **Tangent Plane at P :** $(\mathbf{r} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) = r^2$.
- **Circle of Intersection:** Radius $R = \sqrt{r^2 - d^2}$.
- **Spherical Cosine Rule:** $\cos a = \cos b \cos c + \sin b \sin c \cos A$.

13.0.27 Solid

- **Cone:** $V = \frac{1}{3}\pi r^2 h$, $SA = \pi r(r + l)$
- **Sphere:** $V = \frac{4}{3}\pi r^3$, $SA = 4\pi r^2$
- **Spherical Cap:** $SA = 2\pi rh$. $V = \frac{1}{3}\pi h^2(3r - h)$.
- **Frustum (Cone):** $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$. $LSA = \pi(R + r)l$.
- **Pyramid:** $V = \frac{1}{3}(\text{Base Area}) \times h$. $LSA = \frac{1}{2}(\text{Perimeter}) \times l$.
- **Regular Tetrahedron:** Side a . $H = a\sqrt{2/3}$. $V = \frac{a^3}{6\sqrt{2}}$. Face Angle $\cos \theta = 1/3$.

13.0.28 Trigonometry

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1$
- $\tan^{-1} x \pm \tan^{-1} y = \tan^{-1} \left(\frac{x \pm y}{1 \mp xy} \right)$ (Note: For $+$, if $xy > 1$, add π to the result.)
- $2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1 - x^2} \right) = \sin^{-1} \left(\frac{2x}{1 + x^2} \right) = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right)$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$